

Table 1: Hyper-parameter settings used in the experiments.

Model	Grace period	Ensemble size	Subspaces of features	Drift detector	Background learner
ARF	50	100	$0.6n$	ADWIN	Yes
ARTE	100	100	$\sim \text{Unif}\{2, n\}^1$	ADWIN	No
LB	200	100	n	ADWIN	No
SRP	50	100	$0.6n$	ADWIN	Yes

Table 2: Comparison of the accuracy metric between the ARF algorithm and the PBIS strategy applied at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARF	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	66,65(0,02)	66,50 (0,02)	66,23 (0,02)	65,95 (0,01)	65,65 (0,02)	65,22 (0,00)	64,79 (0,04)	64,21 (0,06)	63,69 (0,04)	62,80 (0,02)	61,78 (0,00)
BankFraud	98,89(0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,85 (0,00)	97,85 (0,00)
Census	95,30 (0,01)	95,33 (0,02)	95,32 (0,00)	95,35 (0,00)	95,36 (0,01)	95,35 (0,03)	95,37 (0,01)	95,34 (0,01)	95,26 (0,01)	95,01 (0,02)	93,91 (0,00)
Coil2000	93,90 (0,00)	93,92 (0,01)	93,88 (0,00)	93,91 (0,02)	93,84 (0,04)	93,76 (0,04)	93,66 (0,03)	93,49 (0,06)	93,41 (0,09)	92,65 (0,01)	88,88 (0,00)
Connect-4	78,01 (0,05)	78,21 (0,03)	78,28 (0,04)	78,53 (0,00)	78,57 (0,00)	78,97 (0,06)	79,08 (0,01)	79,40 (0,02)	79,64 (0,04)	79,65 (0,19)	77,60 (0,00)
CovType	94,71 (0,00)	94,75 (0,00)	94,80 (0,01)	94,85 (0,00)	94,85 (0,01)	94,88 (0,01)	94,85 (0,03)	94,83 (0,01)	94,78 (0,02)	94,72 (0,01)	94,68 (0,00)
Electricity	90,59 (0,06)	90,57 (0,01)	90,55 (0,09)	90,41 (0,00)	90,52 (0,01)	90,45 (0,01)	90,40 (0,09)	90,54 (0,01)	90,52 (0,14)	90,50 (0,19)	90,67 (0,00)
Gassensor	90,47 (0,10)	89,48 (0,30)	89,67 (0,01)	89,60 (0,35)	88,99 (0,78)	88,80 (0,78)	87,62 (0,25)	87,24 (0,29)	87,23 (0,32)	89,04 (0,44)	92,27 (0,00)
GMSC	93,57(0,02)	93,56 (0,01)	93,57 (0,00)	93,53 (0,01)	93,52 (0,02)	93,47 (0,01)	93,39 (0,02)	93,26 (0,01)	93,06 (0,01)	92,63 (0,01)	90,10 (0,00)
Kddcup99	99,86(0,00)	99,88 (0,01)	99,86 (0,01)	99,86 (0,01)	99,88 (0,00)	99,87 (0,02)	99,84 (0,01)	99,86 (0,01)	99,82 (0,01)	99,76 (0,02)	99,63 (0,00)
Keystroke	91,92 (0,17)	92,32 (0,13)	91,75 (0,19)	92,37 (0,00)	91,75 (0,19)	91,75 (0,19)	91,65 (0,53)	91,09 (0,53)	89,81 (0,25)	89,72 (0,53)	88,37 (0,00)
NOAA	79,30(0,03)	79,25 (0,15)	79,32 (0,06)	79,56 (0,04)	79,46 (0,01)	79,39 (0,34)	79,03 (0,09)	78,65 (0,12)	77,92 (0,06)	76,48 (0,20)	74,01 (0,00)
Nomao	97,24 (0,05)	97,23 (0,01)	97,23 (0,05)	97,16 (0,02)	97,04 (0,06)	97,07 (0,01)	97,08 (0,05)	97,01 (0,06)	96,98 (0,04)	97,03 (0,02)	96,96 (0,00)
Outdoor	63,79 (0,54)	64,27 (0,03)	63,48 (0,36)	63,62 (0,22)	63,69 (0,61)	64,25 (0,02)	65,05 (0,52)	65,34 (0,88)	65,14 (0,63)	66,64 (0,73)	67,92 (0,00)
Ozone	93,67 (0,06)	93,71 (0,04)	93,72 (0,08)	93,74 (0,02)	93,84 (0,08)	93,90 (0,14)	94,08 (0,12)	93,80 (0,24)	93,88 (0,08)	93,39 (0,09)	91,16 (0,00)
Poker	88,78 (0,02)	88,95 (0,02)	89,11 (0,02)	89,25 (0,02)	89,46 (0,00)	89,66 (0,00)	89,87 (0,05)	90,24 (0,00)	90,81 (0,04)	91,44 (0,06)	93,12 (0,00)
Rialto	72,21(0,02)	71,85 (0,03)	71,58 (0,27)	70,90 (0,02)	70,37 (0,05)	69,80 (0,25)	69,53 (0,04)	68,38 (0,17)	69,38 (0,14)	70,66 (0,05)	74,33 (0,00)
Shuttle	99,74(0,01)	99,73 (0,02)	99,74 (0,03)	99,72 (0,02)	99,74 (0,00)	99,71 (0,02)	99,69 (0,02)	99,75 (0,02)	99,70 (0,01)	99,69 (0,00)	99,65 (0,00)
AGR	94,07 (0,01)	94,08 (0,00)	94,10 (0,03)	94,07 (0,03)	94,03 (0,04)	94,01 (0,02)	93,93 (0,01)	93,78 (0,04)	93,56 (0,04)	92,91 (0,03)	90,13 (0,00)
AGR_a	87,85 (0,08)	87,90 (0,02)	87,80 (0,14)	87,96 (0,00)	87,66 (0,02)	87,29 (0,22)	86,78 (0,04)	85,87 (0,08)	84,68 (0,23)	83,23 (0,02)	82,49 (0,00)
AGR_g	82,40 (0,05)	82,55 (0,04)	82,58 (0,06)	82,63 (0,11)	82,55 (0,09)	82,15 (0,01)	81,75 (0,20)	81,21 (0,05)	80,53 (0,08)	79,37 (0,22)	78,28 (0,00)
HYPER	85,73 (0,01)	85,94 (0,00)	86,15 (0,02)	86,42 (0,01)	86,69 (0,01)	87,00 (0,01)	87,28 (0,02)	87,68 (0,02)	87,72 (0,03)	86,20 (0,02)	80,82 (0,00)
LED	73,90 (0,01)	73,88 (0,01)	73,89 (0,01)	73,90 (0,01)	73,88 (0,00)	73,88 (0,01)	73,88 (0,02)	73,88 (0,00)	73,84 (0,02)	72,58 (0,02)	68,67 (0,00)
LED_a	73,95 (0,00)	73,96 (0,01)	73,93 (0,01)	73,96 (0,00)	73,96 (0,00)	73,93 (0,00)	73,90 (0,02)	73,84 (0,02)	73,77 (0,02)	72,47 (0,00)	69,00 (0,00)
LED_g	73,12(0,01)	73,14 (0,01)	73,11 (0,01)	73,12 (0,02)	73,14 (0,02)	73,12 (0,01)	73,10 (0,00)	73,01 (0,01)	72,90 (0,01)	71,65 (0,01)	68,20 (0,00)
MIXED	99,71(0,00)	99,71 (0,00)	99,71 (0,00)	99,72 (0,01)	99,72 (0,01)	99,71 (0,01)	99,70 (0,00)	99,71 (0,00)	99,71 (0,00)	99,69 (0,00)	99,42 (0,00)
RBF_f	76,28 (0,00)	75,90 (0,03)	75,48 (0,02)	74,98 (0,01)	74,45 (0,00)	73,83 (0,02)	73,28 (0,02)	72,68 (0,01)	71,99 (0,03)	71,39 (0,02)	70,14 (0,00)
RBF_m	87,17 (0,01)	87,14 (0,02)	87,11 (0,01)	87,08 (0,01)	87,02 (0,00)	86,93 (0,02)	86,80 (0,02)	86,65 (0,02)	86,42 (0,01)	85,88 (0,00)	84,13 (0,00)
RTG	96,09 (0,07)	96,06 (0,07)	96,03 (0,29)	95,96 (0,26)	96,09 (0,24)	96,05 (0,31)	95,99 (0,22)	95,60 (0,18)	96,09 (0,05)	96,11 (0,09)	94,89 (0,00)
SEA	89,67(0,00)	89,65 (0,00)	89,64 (0,01)	89,59 (0,00)	89,53 (0,00)	89,45 (0,00)	89,33 (0,01)	88,98 (0,02)	87,70 (0,01)	84,66 (0,01)	80,24 (0,00)
SEA_a	89,46(0,01)	89,42 (0,02)	89,38 (0,01)	89,31 (0,01)	89,23 (0,02)	89,10 (0,00)	88,94 (0,00)	88,59 (0,01)	87,66 (0,01)	85,48 (0,01)	82,16 (0,00)
SEA_g	88,77(0,35)	88,93 (0,01)	88,91 (0,01)	88,86 (0,01)	88,78 (0,00)	88,69 (0,02)	88,58 (0,02)	88,34 (0,01)	87,52 (0,01)	85,42 (0,02)	81,94 (0,00)
SINE	99,52(0,00)	99,52 (0,00)	99,51 (0,01)	99,50 (0,02)	99,49 (0,00)	99,50 (0,02)	99,46 (0,01)	99,45 (0,00)	99,38 (0,00)	99,30 (0,02)	98,82 (0,00)
WAVEFORM	85,46 (0,03)	85,50 (0,02)	85,55 (0,01)	85,61 (0,05)	85,62 (0,04)	85,58 (0,02)	85,50 (0,02)	85,32 (0,02)	84,86 (0,02)	83,92 (0,02)	81,06 (0,00)
WAVEFORM_g	85,18 (0,00)	85,22 (0,01)	85,22 (0,02)	85,29 (0,04)	85,28 (0,01)	85,20 (0,01)	85,11 (0,02)	84,90 (0,03)	84,53 (0,02)	83,51 (0,03)	80,50 (0,00)
Ranking médio geral	4.06	3.77	4.37	3.97	4.23	5.09	6.37	6.66	7.57	8.54	9.57

Table 3: Comparison of the Kappa M metric between the ARF algorithm and the application of the PBIS strategy at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARF	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	25.14 (0.05)	24.81 (0.04)	24.18 (0.05)	23.55 (0.02)	22.89 (0.04)	21.93 (0.00)	20.96 (0.10)	19.65 (0.14)	18.48 (0.07)	16.49 (0.03)	14.21 (0.00)
BankFraud	-0.02 (0.01)	0.02 (0.01)	0.02 (0.02)	0.00 (0.00)	0.02 (0.02)	0.01 (0.01)	0.02 (0.02)	-0.17 (0.10)	-0.38 (0.07)	-3.64 (0.10)	-94.52 (0.00)
Census	24.34 (0.13)	24.75 (0.17)	24.64 (0.07)	24.99 (0.08)	25.29 (0.18)	25.06 (0.39)	25.40 (0.11)	24.84 (0.05)	23.67 (0.20)	19.52 (0.40)	1.99 (0.00)
Coil2000	-2.17 (0.28)	-1.84 (0.17)	-2.64 (0.09)	-2.13 (0.26)	-3.15 (0.60)	-4.43 (0.68)	-6.23 (0.60)	-9.13 (1.11)	-10.49 (1.62)	-23.03 (0.17)	-86.34 (0.00)
Connect-4	35.66 (0.14)	36.25 (0.09)	36.43 (0.12)	37.18 (0.02)	37.29 (0.00)	38.46 (0.20)	38.77 (0.01)	39.72 (0.05)	40.40 (0.10)	40.44 (0.54)	34.44 (0.00)
CovType	89.68 (0.02)	89.76 (0.01)	89.87 (0.02)	89.96 (0.00)	89.96 (0.02)	90.02 (0.02)	89.96 (0.06)	89.92 (0.02)	89.81 (0.04)	89.71 (0.02)	89.62 (0.00)
Electricity	77.84 (0.13)	77.80 (0.00)	77.73 (0.20)	77.40 (0.01)	77.66 (0.02)	77.51 (0.02)	77.38 (0.21)	77.71 (0.02)	77.68 (0.33)	77.63 (0.44)	78.03 (0.00)
Gassensor	87.55 (0.12)	86.26 (0.38)	86.50 (0.01)	86.43 (0.45)	85.62 (1.02)	85.37 (1.02)	83.84 (0.32)	83.34 (0.38)	83.32 (0.41)	85.69 (0.58)	89.91 (0.00)
GMSC	3.80 (0.22)	3.71 (0.12)	3.76 (0.12)	3.14 (0.07)	3.11 (0.30)	2.41 (0.13)	1.07 (0.27)	-0.72 (0.15)	-3.79 (0.21)	-10.25 (0.13)	-48.04 (0.00)
Kddcup99	99.67 (0.01)	99.72 (0.02)	99.67 (0.02)	99.67 (0.02)	99.72 (0.01)	99.69 (0.03)	99.64 (0.02)	99.68 (0.02)	99.59 (0.02)	99.45 (0.04)	99.15 (0.00)
Keystroke	82.67 (0.34)	83.53 (0.27)	82.30 (0.40)	83.64 (0.00)	82.30 (0.40)	82.30 (0.40)	82.10 (1.14)	80.89 (1.14)	78.15 (0.54)	77.94 (1.14)	75.06 (0.00)
NOAA	34.02 (0.09)	33.88 (0.48)	34.10 (0.20)	34.87 (0.13)	34.55 (0.02)	34.32 (1.07)	33.17 (0.29)	31.97 (0.37)	29.65 (0.21)	25.05 (0.64)	17.19 (0.00)
Nomao	90.36 (0.18)	90.33 (0.00)	90.31 (0.17)	90.06 (0.04)	89.65 (0.21)	89.77 (0.03)	89.79 (0.16)	89.56 (0.21)	89.45 (0.13)	89.60 (0.06)	89.38 (0.00)
Outdoor	62.77 (0.55)	63.25 (0.04)	62.45 (0.37)	62.59 (0.23)	62.66 (0.63)	63.24 (0.02)	64.06 (0.54)	64.35 (0.91)	64.15 (0.65)	65.69 (0.76)	67.01 (0.00)
Ozone	-0.15 (0.81)	0.50 (0.63)	0.63 (1.25)	0.94 (0.32)	2.50 (1.25)	3.44 (2.19)	6.25 (1.88)	1.88 (3.75)	3.12 (1.25)	-4.68 (1.56)	-39.99 (0.00)
Poker	77.51 (0.06)	77.85 (0.05)	78.17 (0.05)	78.45 (0.02)	78.87 (0.01)	79.27 (0.01)	79.69 (0.08)	80.44 (0.00)	81.59 (0.08)	82.84 (0.13)	86.21 (0.00)
Rialto	69.13 (0.16)	68.72 (0.03)	68.42 (0.30)	67.67 (0.02)	67.08 (0.05)	66.44 (0.27)	66.14 (0.04)	64.87 (0.19)	65.98 (0.16)	67.40 (0.05)	71.48 (0.00)
Shuttle	98.79 (0.07)	98.76 (0.09)	98.78 (0.14)	98.70 (0.07)	98.80 (0.02)	98.66 (0.09)	98.55 (0.12)	98.87 (0.09)	98.61 (0.05)	98.54 (0.04)	98.39 (0.00)
AGR	81.92 (0.02)	81.96 (0.02)	82.02 (0.10)	81.94 (0.09)	81.80 (0.11)	81.75 (0.05)	81.50 (0.02)	81.02 (0.11)	80.35 (0.13)	78.38 (0.08)	69.92 (0.00)
AGR_a	74.24 (0.17)	74.36 (0.05)	74.13 (0.28)	74.47 (0.02)	73.85 (0.05)	73.05 (0.47)	71.97 (0.08)	70.05 (0.16)	67.53 (0.48)	64.46 (0.04)	62.89 (0.00)
AGR_g	62.69 (0.13)	63.01 (0.09)	63.08 (0.13)	63.19 (0.23)	63.01 (0.18)	62.16 (0.00)	61.31 (0.44)	60.17 (0.11)	58.74 (0.16)	56.27 (0.46)	53.95 (0.00)
HYPER	71.42 (0.04)	71.84 (0.00)	72.25 (0.04)	72.78 (0.01)	73.34 (0.02)	73.96 (0.02)	74.52 (0.03)	75.31 (0.04)	75.39 (0.07)	72.35 (0.05)	61.57 (0.00)
LED	70.98 (0.01)	70.96 (0.01)	70.96 (0.00)	70.97 (0.00)	70.95 (0.00)	70.96 (0.01)	70.95 (0.02)	70.96 (0.00)	70.90 (0.02)	69.51 (0.02)	65.16 (0.00)
LED_a	71.01 (0.00)	71.02 (0.01)	70.99 (0.01)	71.01 (0.01)	71.02 (0.00)	70.99 (0.00)	70.95 (0.01)	70.89 (0.02)	70.81 (0.02)	69.37 (0.01)	65.50 (0.00)
LED_g	70.08 (0.02)	70.10 (0.00)	70.08 (0.02)	70.08 (0.02)	70.11 (0.02)	70.08 (0.01)	70.06 (0.01)	69.95 (0.01)	69.84 (0.01)	68.44 (0.01)	64.61 (0.00)
MIXED	99.06 (0.00)	99.04 (0.01)	99.02 (0.02)	99.07 (0.00)	99.05 (0.02)	99.05 (0.02)	99.02 (0.00)	99.01 (0.02)	99.03 (0.02)	98.95 (0.01)	98.09 (0.00)
RBF_f	66.11 (0.04)	65.56 (0.05)	64.97 (0.03)	64.25 (0.01)	63.50 (0.00)	62.61 (0.03)	61.82 (0.03)	60.96 (0.00)	59.98 (0.05)	59.12 (0.02)	57.34 (0.00)
RBF_m	81.67 (0.02)	81.62 (0.02)	81.59 (0.02)	81.54 (0.01)	81.45 (0.01)	81.33 (0.02)	81.14 (0.02)	80.93 (0.03)	80.59 (0.01)	79.83 (0.00)	77.33 (0.00)
RTG	90.73 (0.13)	90.65 (0.17)	90.58 (0.68)	90.44 (0.62)	90.72 (0.58)	90.64 (0.73)	90.49 (0.55)	89.57 (0.42)	90.73 (0.13)	90.77 (0.22)	87.88 (0.00)
SEA	71.06 (0.01)	71.01 (0.00)	70.95 (0.01)	70.83 (0.01)	70.66 (0.01)	70.44 (0.00)	70.10 (0.02)	69.12 (0.07)	65.53 (0.02)	57.00 (0.03)	44.61 (0.00)
SEA_a	75.58 (0.02)	75.49 (0.03)	75.39 (0.02)	75.24 (0.03)	75.04 (0.04)	74.74 (0.02)	74.36 (0.01)	73.56 (0.03)	71.40 (0.02)	66.34 (0.02)	58.66 (0.00)
SEA_g	73.46 (1.65)	74.35 (0.02)	74.31 (0.02)	74.17 (0.01)	74.00 (0.01)	73.79 (0.05)	73.54 (0.05)	72.97 (0.01)	71.08 (0.02)	66.22 (0.05)	58.14 (0.00)
SINE	98.97 (0.01)	98.96 (0.01)	98.93 (0.00)	98.91 (0.03)	98.89 (0.01)	98.91 (0.03)	98.83 (0.01)	98.81 (0.01)	98.66 (0.00)	98.47 (0.03)	97.43 (0.00)
WAVEFORM	78.14 (0.04)	78.21 (0.02)	78.29 (0.02)	78.38 (0.08)	78.39 (0.05)	78.32 (0.04)	78.21 (0.03)	77.94 (0.02)	77.24 (0.04)	75.83 (0.04)	71.54 (0.00)
WAVEFORM_g	77.74 (0.00)	77.79 (0.02)	77.79 (0.04)	77.90 (0.05)	77.89 (0.01)	77.75 (0.02)	77.63 (0.03)	77.32 (0.05)	76.75 (0.02)	75.22 (0.04)	70.71 (0.00)
Ranking médio geral	4.26	4.03	4.51	4.20	4.40	5.20	6.31	6.94	7.74	8.57	9.57

Table 4: Comparison of the accuracy metric between the ARTE algorithm and the PBIS strategy applied at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARTE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	67.45 (0.18)	65.88 (0.02)	63.39 (0.02)	63.12 (0.05)	62.88 (0.04)	62.59 (0.02)	62.14 (0.01)	61.67 (0.07)	61.10 (0.05)	60.29 (0.05)	59.56 (0.00)
BankFraud	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.88 (0.00)	98.82 (0.00)	97.80 (0.00)
Census	95.69 (0.05)	95.78 (0.01)	95.36 (0.02)	95.37 (0.01)	95.37 (0.02)	95.36 (0.01)	95.34 (0.01)	95.27 (0.00)	95.13 (0.03)	94.91 (0.00)	93.97 (0.00)
Coil2000	93.94 (0.05)	93.89 (0.03)	93.89 (0.02)	93.89 (0.02)	93.91 (0.00)	93.77 (0.03)	93.83 (0.04)	93.56 (0.03)	93.32 (0.09)	92.38 (0.11)	88.49 (0.00)
Connect-4	78.30 (0.36)	78.19 (0.05)	78.31 (0.02)	78.42 (0.06)	78.62 (0.02)	78.73 (0.07)	78.97 (0.00)	79.16 (0.05)	79.41 (0.00)	79.37 (0.01)	77.29 (0.00)
CovType	94.96 (0.07)	95.28 (0.02)	95.33 (0.01)	95.34 (0.02)	95.36 (0.01)	95.29 (0.01)	95.30 (0.00)	95.28 (0.02)	95.19 (0.02)	95.09 (0.00)	94.71 (0.00)
Electricity	91.13 (0.23)	90.71 (0.07)	90.71 (0.07)	90.74 (0.07)	90.71 (0.10)	90.74 (0.05)	90.72 (0.07)	90.72 (0.15)	90.87 (0.05)	90.80 (0.19)	90.15 (0.00)
Gassensor	95.58 (0.22)	95.63 (0.10)	95.67 (0.02)	95.44 (0.00)	95.59 (0.15)	95.28 (0.04)	95.23 (0.14)	95.05 (0.07)	95.14 (0.02)	95.16 (0.20)	95.84 (0.00)
GMSC	93.38 (0.02)	93.57 (0.01)	93.56 (0.01)	93.55 (0.00)	93.51 (0.01)	93.46 (0.02)	93.41 (0.03)	93.31 (0.02)	93.09 (0.02)	92.61 (0.04)	89.90 (0.00)
Kddcup99	99.78 (0.04)	99.89 (0.00)	99.90 (0.01)	99.89 (0.00)	99.89 (0.00)	99.90 (0.01)	99.89 (0.00)	99.89 (0.00)	99.88 (0.00)	99.89 (0.01)	99.86 (0.00)
Keystroke	94.04 (0.29)	93.71 (0.18)	93.47 (0.09)	93.81 (0.25)	93.47 (0.09)	93.63 (0.13)	92.59 (0.47)	92.56 (0.25)	91.65 (0.03)	90.62 (0.75)	88.00 (0.00)
NOAA	79.08 (0.00)	79.43 (0.13)	79.44 (0.05)	79.63 (0.03)	79.62 (0.00)	79.49 (0.04)	79.08 (0.24)	78.78 (0.21)	77.92 (0.04)	76.57 (0.20)	74.26 (0.00)
Nomao	97.40 (0.04)	97.37 (0.04)	97.30 (0.01)	97.27 (0.07)	97.33 (0.02)	97.24 (0.03)	97.29 (0.01)	97.26 (0.04)	97.27 (0.12)	97.22 (0.02)	96.97 (0.00)
Outdoor	71.97 (1.23)	73.08 (0.19)	73.00 (0.40)	72.89 (0.09)	72.42 (0.55)	72.01 (0.12)	73.05 (0.48)	72.87 (0.80)	72.44 (0.72)	73.57 (0.00)	75.82 (0.00)
Ozone	93.83 (0.13)	93.90 (0.14)	93.94 (0.06)	93.88 (0.04)	93.88 (0.04)	93.96 (0.04)	93.96 (0.08)	94.06 (0.05)	93.70 (0.14)	93.27 (0.22)	91.43 (0.00)
Poker	94.43 (1.42)	93.21 (0.02)	87.22 (0.05)	87.19 (0.00)	87.05 (0.06)	86.94 (0.02)	86.82 (0.08)	86.70 (0.01)	86.75 (0.02)	87.33 (0.06)	88.73 (0.00)
Rialto	80.78 (0.40)	79.09 (0.08)	79.36 (0.17)	78.90 (0.04)	78.26 (0.04)	77.84 (0.07)	77.34 (0.02)	76.84 (0.02)	76.31 (0.07)	76.05 (0.03)	79.19 (0.00)
Shuttle	99.74 (0.01)	99.73 (0.01)	99.72 (0.01)	99.73 (0.00)	99.70 (0.01)	99.68 (0.00)	99.66 (0.00)	99.64 (0.02)	99.53 (0.04)	99.08 (0.02)	97.21 (0.00)
AGR	94.99 (0.04)	94.96 (0.00)	93.16 (0.03)	93.37 (0.02)	93.32 (0.05)	93.34 (0.01)	93.33 (0.06)	93.00 (0.05)	92.61 (0.05)	91.17 (0.12)	88.09 (0.00)
AGR_a	94.40 (0.02)	94.35 (0.01)	79.95 (0.07)	80.26 (0.06)	80.19 (0.02)	80.21 (0.02)	79.97 (0.01)	79.31 (0.07)	78.63 (0.08)	77.52 (0.01)	77.93 (0.00)
AGR_g	91.86 (0.02)	91.71 (0.02)	76.01 (0.09)	76.17 (0.00)	76.05 (0.02)	75.80 (0.11)	75.50 (0.16)	74.97 (0.07)	74.49 (0.02)	73.71 (0.13)	73.37 (0.00)
HYPER	87.52 (0.19)	87.07 (0.03)	87.27 (0.02)	87.45 (0.01)	87.66 (0.00)	87.95 (0.01)	88.22 (0.01)	88.44 (0.00)	88.21 (0.01)	86.06 (0.07)	80.16 (0.00)
LED	73.50 (0.71)	73.91 (29.56)	73.91 (0.01)	73.92 (0.01)	73.90 (0.01)	73.93 (0.02)	73.91 (0.00)	73.91 (0.00)	73.88 (0.02)	72.70 (0.01)	68.78 (0.00)
LED_a	73.04 (0.93)	73.91 (0.02)	73.93 (0.02)	73.89 (0.01)	73.92 (0.01)	73.92 (0.02)	73.91 (0.00)	73.86 (0.00)	73.75 (0.02)	72.56 (0.04)	68.95 (0.00)
LED_g	71.93 (1.03)	72.98 (0.01)	73.01 (0.02)	73.04 (0.01)	73.04 (0.01)	73.07 (0.02)	73.03 (0.02)	72.97 (0.01)	72.76 (0.01)	71.60 (0.03)	68.08 (0.00)
MIXED	99.78 (0.00)	99.78 (0.00)	99.78 (0.01)	99.78 (0.01)	99.77 (0.00)	99.76 (0.00)	99.76 (0.01)	99.75 (0.01)	99.71 (0.00)	99.56 (0.00)	97.62 (0.00)
RBF_f	79.27 (0.34)	79.41 (0.02)	79.14 (0.01)	78.72 (0.01)	78.14 (0.02)	77.36 (0.02)	76.27 (0.02)	74.91 (0.01)	73.46 (0.02)	72.14 (0.02)	70.76 (0.00)
RBF_m	87.55 (0.08)	87.65 (0.02)	87.62 (0.02)	87.62 (0.01)	87.62 (0.01)	87.59 (0.01)	87.53 (0.01)	87.48 (0.01)	87.28 (0.02)	86.62 (0.02)	84.20 (0.00)
RTG	96.89 (0.29)	96.81 (0.07)	93.62 (0.15)	93.47 (0.08)	93.72 (0.02)	93.99 (0.14)	93.73 (0.29)	93.69 (0.34)	93.73 (0.04)	93.39 (0.02)	91.97 (0.00)
SEA	89.87 (0.00)	89.87 (0.00)	89.85 (0.00)	89.84 (0.01)	89.83 (0.01)	89.81 (0.01)	89.76 (0.00)	89.69 (0.00)	89.41 (0.00)	86.30 (0.03)	80.66 (0.00)
SEA_a	89.73 (0.01)	89.73 (0.00)	89.71 (0.00)	89.69 (0.00)	89.65 (0.00)	89.61 (0.01)	89.53 (0.00)	89.40 (0.02)	88.99 (0.03)	86.04 (0.02)	80.72 (0.00)
SEA_g	89.31 (0.01)	89.30 (0.00)	89.30 (0.00)	89.28 (0.01)	89.25 (0.01)	89.21 (0.00)	89.14 (0.00)	89.01 (0.01)	88.57 (0.02)	85.64 (0.05)	80.32 (0.00)
SINE	99.88 (0.00)	99.87 (0.00)	99.87 (0.01)	99.87 (0.01)	99.86 (0.00)	99.86 (0.01)	99.85 (0.00)	99.84 (0.01)	99.81 (0.00)	99.77 (0.00)	99.43 (0.00)
WAVEFORM	85.70 (0.05)	85.91 (0.00)	85.92 (0.00)	85.93 (0.02)	85.91 (0.00)	85.85 (0.01)	85.72 (0.01)	85.45 (0.01)	84.91 (0.00)	83.83 (0.00)	80.74 (0.00)
WAVEFORM_g	85.39 (0.06)	85.67 (0.01)	85.70 (0.01)	85.69 (0.01)	85.67 (0.01)	85.60 (0.00)	85.48 (0.02)	85.19 (0.00)	84.66 (0.03)	83.63 (0.05)	80.27 (0.00)
Ranking médio geral	3.82	3.82	3.76	3.47	4.35	4.41	5.65	7.18	8.18	10.00	10.94

Table 5: Comparison of the Kappa M metric between the ARTE algorithm and the application of the PBIS strategy at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARTE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	26.94 (0.40)	23.41 (0.05)	17.80 (0.03)	17.21 (0.12)	16.67 (0.08)	16.01 (0.03)	15.02 (0.02)	13.96 (0.15)	12.66 (0.11)	10.86 (0.11)	9.22 (0.00)
BankFraud	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.01 (0.01)	-0.03 (0.02)	-0.10 (0.01)	-1.00 (0.06)	-6.27 (0.06)	-99.09 (0.00)
Census	30.64 (0.87)	32.07 (0.17)	25.21 (0.19)	25.36 (0.03)	25.40 (0.21)	25.29 (0.14)	24.96 (0.12)	23.78 (0.13)	21.53 (0.51)	17.93 (0.07)	2.82 (0.00)
Coil2000	-1.50 (0.89)	-2.45 (0.53)	-2.78 (0.26)	-2.67 (0.43)	-2.21 (0.09)	-4.43 (0.59)	-3.75 (0.59)	-7.84 (0.52)	-11.66 (1.45)	-26.96 (1.88)	-92.83 (0.00)
Connect-4	36.51 (1.05)	36.19 (0.13)	36.52 (0.08)	36.90 (0.18)	37.48 (0.08)	37.91 (0.22)	38.53 (0.00)	38.94 (0.13)	39.72 (0.00)	39.50 (0.00)	33.53 (0.00)
CovType	90.16 (0.13)	90.79 (0.05)	90.88 (0.01)	90.90 (0.03)	90.94 (0.00)	90.81 (0.01)	90.83 (0.01)	90.78 (0.03)	90.62 (0.05)	90.42 (0.00)	89.67 (0.00)
Electricity	79.10 (0.53)	78.12 (0.17)	78.13 (0.16)	78.19 (0.16)	78.13 (0.24)	78.18 (0.11)	78.15 (0.16)	78.15 (0.35)	78.49 (0.13)	78.33 (0.44)	76.81 (0.00)
Gassensor	94.24 (0.28)	94.30 (0.13)	94.37 (0.02)	94.03 (0.00)	94.15 (0.20)	93.84 (0.05)	93.70 (0.18)	93.59 (0.09)	93.53 (0.04)	93.66 (0.26)	94.57 (0.00)
GMSC	1.00 (0.20)	3.77 (0.23)	3.59 (0.09)	3.49 (0.10)	2.90 (0.04)	2.16 (0.25)	1.38 (0.34)	-0.03 (0.32)	-3.45 (0.24)	-10.58 (0.50)	-51.04 (0.00)
Kddcup99	99.48 (0.09)	99.76 (0.01)	99.77 (0.00)	99.75 (0.00)	99.76 (0.01)	99.76 (0.01)	99.75 (0.00)	99.76 (0.01)	99.74 (0.00)	99.74 (0.00)	99.67 (0.00)
Keystroke	87.21 (0.61)	86.51 (0.37)	85.97 (0.20)	85.88 (0.54)	86.28 (0.20)	85.96 (0.27)	83.86 (1.01)	84.31 (0.53)	82.39 (0.07)	80.02 (1.61)	74.26 (0.00)
NOAA	33.34 (0.01)	34.45 (0.42)	34.67 (0.14)	35.25 (0.08)	35.23 (0.00)	34.63 (0.12)	33.51 (0.78)	32.33 (0.67)	29.63 (0.10)	25.12 (0.62)	17.97 (0.00)
Nomao	90.90 (0.14)	90.79 (0.13)	90.67 (0.03)	90.51 (0.24)	90.63 (0.08)	90.37 (0.10)	90.53 (0.03)	90.37 (0.14)	90.33 (0.42)	90.34 (0.05)	89.42 (0.00)
Outdoor	71.18 (1.27)	72.31 (0.20)	72.11 (0.41)	72.04 (0.09)	71.25 (0.57)	71.43 (0.12)	72.08 (0.49)	72.53 (0.83)	72.06 (0.73)	72.92 (0.00)	75.14 (0.00)
Ozone	2.37 (2.06)	3.37 (2.11)	4.06 (0.94)	3.13 (0.63)	3.13 (0.63)	4.38 (0.63)	4.37 (1.25)	5.94 (0.93)	0.32 (2.19)	-6.56 (3.44)	-35.62 (0.00)
Poker	88.84 (2.85)	86.39 (0.05)	74.38 (0.09)	74.32 (0.00)	74.05 (0.11)	73.83 (0.04)	73.59 (0.16)	73.34 (0.01)	73.44 (0.05)	74.60 (0.13)	77.42 (0.00)
Rialto	78.64 (0.44)	76.76 (0.09)	77.06 (0.20)	76.55 (0.04)	75.84 (0.04)	75.37 (0.07)	74.82 (0.03)	74.26 (0.03)	73.68 (0.08)	73.39 (0.03)	76.88 (0.00)
Shuttle	98.81 (0.04)	98.77 (0.04)	98.71 (0.05)	98.75 (0.01)	98.64 (0.05)	98.53 (0.01)	98.40 (0.02)	98.34 (0.09)	97.83 (0.17)	95.73 (0.09)	86.97 (0.00)
AGR	84.73 (0.12)	84.63 (0.02)	79.14 (0.09)	79.79 (0.03)	79.64 (0.16)	79.69 (0.01)	79.65 (0.19)	78.66 (0.17)	77.47 (0.13)	73.09 (0.35)	63.71 (0.00)
AGR_a	88.13 (0.04)	88.01 (0.02)	57.49 (0.14)	58.14 (0.14)	58.01 (0.04)	58.04 (0.05)	57.54 (0.03)	56.15 (0.15)	54.69 (0.16)	52.35 (0.01)	53.21 (0.00)
AGR_g	82.75 (0.05)	82.43 (0.05)	49.16 (0.19)	49.49 (0.01)	49.23 (0.05)	48.70 (0.23)	48.07 (0.34)	46.95 (0.15)	45.93 (0.04)	44.28 (0.29)	43.56 (0.00)
HYPER	75.00 (0.39)	74.10 (0.06)	74.50 (0.04)	74.85 (0.01)	75.28 (0.01)	75.87 (0.02)	76.40 (0.01)	76.85 (0.00)	76.39 (0.02)	72.08 (0.13)	60.26 (0.00)
LED	70.53 (0.79)	70.99 (28.40)	70.99 (0.01)	71.00 (0.02)	70.98 (0.01)	71.00 (0.02)	70.98 (0.01)	70.98 (0.01)	70.95 (0.02)	69.64 (0.01)	65.28 (0.00)
LED_a	70.00 (1.04)	70.97 (0.01)	70.98 (0.03)	70.94 (0.00)	70.98 (0.02)	70.97 (0.01)	70.97 (0.01)	70.91 (0.00)	70.79 (0.02)	69.45 (0.04)	65.44 (0.00)
LED_g	68.76 (1.15)	69.92 (0.01)	69.96 (0.02)	69.99 (0.00)	69.99 (0.01)	70.02 (0.03)	69.98 (0.02)	69.91 (0.01)	69.68 (0.01)	68.39 (0.04)	64.47 (0.00)
MIXED	99.29 (0.02)	99.28 (0.01)	99.27 (0.02)	99.26 (0.00)	99.25 (0.01)	99.23 (0.01)	99.19 (0.01)	99.16 (0.02)	99.03 (0.01)	98.54 (0.01)	92.07 (0.00)
RBF_f	70.38 (0.49)	70.59 (0.03)	70.20 (0.02)	69.61 (0.02)	68.76 (0.02)	67.64 (0.02)	66.09 (0.02)	64.15 (0.02)	62.08 (0.04)	60.20 (0.04)	58.22 (0.00)
RBF_m	82.21 (0.11)	82.35 (0.03)	82.32 (0.03)	82.32 (0.02)	82.32 (0.02)	82.27 (0.02)	82.19 (0.02)	82.11 (0.01)	81.83 (0.02)	80.89 (0.02)	77.43 (0.00)
RTG	92.63 (0.70)	92.45 (0.17)	84.87 (0.34)	84.52 (0.20)	85.09 (0.04)	85.74 (0.33)	85.14 (0.68)	85.03 (0.79)	85.12 (0.09)	84.32 (0.05)	80.95 (0.00)
SEA	71.60 (0.01)	71.60 (0.01)	71.57 (0.00)	71.53 (0.02)	71.49 (0.01)	71.44 (0.00)	71.31 (0.00)	71.09 (0.02)	70.31 (0.02)	61.62 (0.07)	45.81 (0.00)
SEA_a	76.20 (0.01)	76.20 (0.01)	76.15 (0.02)	76.10 (0.00)	76.03 (0.01)	75.93 (0.02)	75.74 (0.00)	75.43 (0.04)	74.48 (0.05)	67.65 (0.04)	55.33 (0.00)
SEA_g	75.23 (0.02)	75.21 (0.01)	75.19 (0.01)	75.15 (0.01)	75.08 (0.00)	75.00 (0.01)	74.82 (0.00)	74.52 (0.00)	73.52 (0.05)	66.71 (0.10)	54.38 (0.00)
SINE	99.74 (0.00)	99.73 (0.00)	99.71 (0.00)	99.71 (0.00)	99.70 (0.00)	99.69 (0.00)	99.67 (0.01)	99.65 (0.00)	99.59 (0.00)	99.51 (0.01)	98.77 (0.00)
WAVEFORM	78.51 (0.08)	78.82 (0.01)	78.85 (0.00)	78.85 (0.02)	78.82 (0.01)	78.73 (0.01)	78.54 (0.01)	78.13 (0.02)	77.31 (0.01)	75.70 (0.00)	71.06 (0.00)
WAVEFORM_g	78.05 (0.09)	78.47 (0.01)	78.52 (0.01)	78.51 (0.02)	78.47 (0.01)	78.37 (0.01)	78.19 (0.02)	77.74 (0.01)	76.96 (0.05)	75.41 (0.07)	70.36 (0.00)
Ranking médio geral	3.82	3.76	3.76	3.65	4.24	4.41	5.71	7.18	8.18	10.00	10.94

Table 6: Comparison of the accuracy metric between the SRP algorithm and the PBIS strategy applied at the *ensemble* level. Bold values represent the best results per metric.

Dataset	SRP	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	68.56 (0.00)	68.41 (0.02)	68.11 (0.02)	67.88 (0.05)	67.56 (0.02)	67.09 (0.02)	66.45 (0.02)	65.75 (0.02)	64.91 (0.01)	63.74 (0.01)	62.44 (0.00)
BankFraud	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.89 (0.00)	98.88 (0.00)	97.87 (0.00)
Census	95.27 (0.00)	95.27 (0.01)	95.30 (0.00)	95.32 (0.00)	95.32 (0.00)	95.33 (0.01)	95.31 (0.01)	95.24 (0.01)	95.12 (0.01)	94.79 (0.02)	93.64 (0.00)
Coil2000	93.95 (0.02)	93.94 (0.01)	93.94 (0.02)	93.89 (0.02)	93.89 (0.00)	93.87 (0.02)	93.78 (0.01)	93.69 (0.02)	93.51 (0.01)	92.72 (0.07)	88.91 (0.00)
Connect-4	78.75 (0.06)	78.90 (0.04)	78.98 (0.13)	79.27 (0.03)	79.47 (0.15)	79.57 (0.09)	79.74 (0.11)	79.88 (0.03)	79.80 (0.09)	79.62 (0.13)	78.09 (0.00)
CovType	95.33 (0.02)	95.30 (0.02)	95.27 (0.00)	95.24 (0.00)	95.19 (0.01)	95.12 (0.02)	95.05 (0.00)	94.93 (0.01)	94.85 (0.02)	94.68 (0.01)	94.46 (0.00)
Electricity	89.81 (0.06)	89.57 (0.10)	89.68 (0.05)	89.66 (0.02)	89.45 (0.01)	89.34 (0.13)	89.43 (0.23)	89.24 (0.17)	89.88 (0.11)	90.23 (0.03)	90.76 (0.00)
Gassensor	93.33 (0.10)	93.43 (0.14)	92.74 (0.16)	92.39 (0.37)	92.56 (0.38)	91.58 (0.23)	91.91 (0.07)	92.33 (0.34)	92.62 (0.09)	91.94 (0.69)	92.20 (0.00)
GMSC	93.49 (0.01)	93.52 (0.02)	93.54 (0.01)	93.57 (0.02)	93.59 (0.00)	93.58 (0.02)	93.56 (0.01)	93.52 (0.02)	93.37 (0.02)	92.91 (0.05)	90.13 (0.00)
Kddcup99	99.89 (0.00)	99.89 (0.00)	99.89 (0.01)	99.89 (0.01)	99.89 (0.01)	99.88 (0.00)	99.89 (0.01)	99.89 (0.01)	99.88 (0.00)	99.88 (0.00)	99.88 (0.00)
Keystroke	94.57 (0.19)	94.34 (0.20)	94.34 (0.28)	93.81 (0.00)	94.31 (0.06)	93.00 (0.38)	93.41 (0.09)	92.25 (0.00)	91.63 (0.38)	91.22 (0.16)	89.25 (0.00)
NOAA	79.23 (0.00)	79.53 (0.11)	79.54 (0.09)	79.61 (0.01)	79.59 (0.01)	79.61 (0.31)	79.30 (0.25)	78.96 (0.00)	78.39 (0.07)	76.55 (0.13)	73.93 (0.00)
Nomao	97.38 (0.00)	97.32 (0.03)	97.31 (0.07)	97.27 (0.00)	97.25 (0.05)	97.19 (0.02)	97.23 (0.02)	97.17 (0.02)	97.14 (0.02)	97.17 (0.02)	96.95 (0.00)
Outdoor	68.46 (0.31)	68.61 (0.32)	68.74 (0.27)	69.04 (0.88)	68.65 (0.30)	67.50 (0.48)	68.81 (0.38)	68.03 (0.13)	68.16 (0.41)	68.65 (0.38)	70.30 (0.00)
Ozone	93.64 (0.07)	93.66 (0.11)	93.82 (0.02)	93.92 (0.00)	93.94 (0.02)	93.96 (0.08)	93.90 (0.10)	93.92 (0.08)	93.66 (0.22)	93.11 (0.34)	91.59 (0.00)
Poker	89.75 (0.05)	89.80 (0.04)	89.93 (0.10)	90.04 (0.07)	90.25 (0.07)	90.36 (0.13)	90.37 (0.31)	90.66 (0.08)	91.27 (0.16)	91.65 (0.15)	94.69 (0.00)
Rialto	80.11 (0.11)	79.75 (0.06)	79.25 (0.09)	79.00 (0.09)	78.60 (0.02)	78.08 (0.10)	77.50 (0.05)	77.17 (0.15)	76.44 (0.06)	76.22 (0.10)	77.36 (0.00)
Shuttle	99.74 (0.01)	99.73 (0.01)	99.72 (0.01)	99.72 (0.01)	99.73 (0.02)	99.72 (0.01)	99.71 (0.02)	99.72 (0.00)	99.71 (0.01)	99.67 (0.02)	99.63 (0.00)
AGR	94.67 (0.03)	94.63 (0.03)	94.49 (0.05)	94.57 (0.00)	94.47 (0.03)	94.37 (0.03)	94.24 (0.01)	94.09 (0.04)	93.91 (0.00)	93.56 (0.04)	92.78 (0.00)
AGR_a	92.81 (0.11)	92.94 (0.08)	93.07 (0.22)	92.65 (0.23)	92.82 (0.19)	92.84 (0.02)	92.90 (0.01)	92.61 (0.09)	92.49 (0.06)	92.12 (0.00)	90.47 (0.00)
AGR_g	89.62 (0.14)	89.92 (0.25)	89.87 (0.27)	89.76 (0.14)	89.75 (0.14)	89.66 (0.07)	89.76 (0.06)	89.51 (0.01)	89.20 (0.11)	88.24 (0.03)	86.69 (0.00)
HYPER	85.49 (0.03)	85.66 (0.10)	85.77 (0.01)	85.91 (0.00)	86.08 (0.05)	86.20 (0.01)	86.10 (0.06)	85.96 (0.09)	85.33 (0.05)	83.74 (0.03)	79.39 (0.00)
LED	73.96 (0.01)	73.94 (0.01)	73.93 (0.03)	73.93 (0.01)	73.95 (0.01)	73.95 (0.01)	73.89 (0.01)	73.86 (0.01)	73.36 (0.02)	71.71 (0.03)	69.05 (0.00)
LED_a	74.03 (0.01)	74.03 (0.02)	74.03 (0.02)	74.02 (0.00)	73.99 (0.00)	73.96 (0.00)	73.95 (0.04)	73.82 (0.02)	73.16 (0.00)	71.37 (0.05)	68.80 (0.00)
LED_g	73.24 (0.00)	73.24 (0.01)	73.24 (0.02)	73.24 (0.03)	73.19 (0.02)	73.20 (0.02)	73.12 (0.03)	73.00 (0.02)	72.48 (0.02)	70.79 (0.04)	67.95 (0.00)
MIXED	84.14 (0.01)	85.09 (0.01)	86.20 (0.01)	87.40 (0.02)	88.74 (0.02)	90.13 (0.02)	91.65 (0.01)	93.25 (0.01)	94.93 (0.02)	96.70 (0.01)	98.96 (0.00)
RBF_f	76.37 (0.03)	76.23 (0.03)	76.12 (0.00)	76.00 (0.01)	75.78 (0.02)	75.55 (0.00)	75.32 (0.02)	74.99 (0.00)	74.45 (0.02)	73.50 (0.01)	71.69 (0.00)
RBF_m	86.02 (0.00)	85.98 (0.01)	85.92 (0.02)	85.85 (0.00)	85.82 (0.03)	85.70 (0.01)	85.60 (0.03)	85.54 (0.02)	85.39 (0.02)	85.21 (0.00)	83.99 (0.00)
RTG	92.07 (0.43)	92.10 (0.35)	92.31 (0.43)	92.69 (0.10)	93.46 (0.36)	94.15 (0.01)	94.06 (0.10)	94.62 (0.13)	94.89 (0.02)	95.42 (0.01)	94.75 (0.00)
SEA	89.77 (0.00)	89.76 (0.00)	89.75 (0.00)	89.74 (0.00)	89.71 (0.00)	89.68 (0.01)	89.60 (0.01)	89.40 (0.01)	88.60 (0.02)	85.44 (0.05)	80.50 (0.00)
SEA_a	88.63 (0.16)	88.88 (0.20)	88.53 (0.07)	88.59 (0.17)	88.33 (0.02)	87.69 (0.55)	87.37 (0.07)	85.71 (0.28)	84.95 (0.22)	84.44 (0.11)	80.59 (0.00)
SEA_g	87.25 (0.03)	87.34 (0.14)	87.51 (0.26)	87.46 (0.07)	87.48 (0.27)	87.03 (0.21)	86.26 (0.07)	85.98 (0.25)	86.62 (0.04)	84.89 (0.05)	80.35 (0.00)
SINE	99.74 (0.00)	99.74 (0.00)	99.74 (0.00)	99.73 (0.01)	99.72 (0.01)	99.72 (0.01)	99.70 (0.01)	99.70 (0.01)	99.68 (0.01)	99.64 (0.01)	99.50 (0.00)
WAVEFORM	85.82 (0.00)	85.83 (0.01)	85.82 (0.00)	85.83 (0.01)	85.83 (0.02)	85.74 (0.02)	85.66 (0.00)	85.43 (0.02)	84.93 (0.01)	83.97 (0.00)	81.02 (0.00)
WAVEFORM_g	85.47 (0.00)	85.48 (0.02)	85.49 (0.00)	85.47 (0.01)	85.47 (0.02)	85.39 (0.01)	85.23 (0.04)	85.04 (0.02)	84.54 (0.01)	83.41 (0.10)	80.43 (0.00)
Ranking médio geral	4,09	3,71	3,97	4,17	4,34	5,40	5,89	6,86	7,63	8,69	9,37

Table 7: Comparison of the Kappa M metric between the SRP algorithm and the application of the PBIS strategy at the *ensemble* level. Bold values represent the best results per metric.

Dataset	SRP	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	29.42 (0.00)	29.08 (0.06)	28.40 (0.06)	27.90 (0.11)	27.17 (0.06)	26.12 (0.04)	24.67 (0.06)	23.11 (0.06)	21.24 (0.03)	18.61 (0.03)	15.68 (0.00)
BankFraud	0.00 (0.00)	0.00 (0.00)	0.01 (0.01)	0.00 (0.00)	0.00 (0.00)	-0.02 (0.02)	-0.02 (0.03)	0.06 (0.00)	0.08 (0.12)	-1.24 (0.08)	-93.05 (0.00)
Census	23.80 (0.07)	23.87 (0.22)	24.28 (0.10)	24.57 (0.12)	24.60 (0.11)	24.84 (0.13)	24.38 (0.18)	23.42 (0.15)	21.40 (0.16)	16.00 (0.44)	-2.36 (0.00)
Coil2000	-1.45 (0.28)	-1.50 (0.17)	-1.62 (0.43)	-2.47 (0.43)	-2.38 (0.00)	-2.73 (0.35)	-4.18 (0.09)	-5.63 (0.34)	-8.70 (0.17)	-21.92 (1.11)	-85.83 (0.00)
Connect-4	37.82 (0.17)	38.25 (0.13)	38.50 (0.36)	39.34 (0.09)	39.93 (0.45)	40.22 (0.27)	40.72 (0.33)	41.13 (0.09)	40.90 (0.26)	40.38 (0.38)	35.88 (0.00)
CovType	90.89 (0.05)	90.84 (0.04)	90.77 (0.01)	90.72 (0.00)	90.62 (0.02)	90.48 (0.04)	90.34 (0.02)	90.12 (0.02)	89.94 (0.03)	89.62 (0.01)	89.19 (0.00)
Electricity	75.99 (0.14)	75.43 (0.23)	75.69 (0.13)	75.64 (0.05)	75.14 (0.00)	74.88 (0.32)	75.09 (0.55)	74.64 (0.38)	76.17 (0.26)	77.00 (0.07)	78.24 (0.00)
Gassensor	91.29 (0.13)	91.42 (0.19)	90.51 (0.20)	90.05 (0.48)	90.28 (0.51)	89.01 (0.30)	89.44 (0.09)	89.98 (0.44)	90.35 (0.12)	89.48 (0.91)	89.82 (0.00)
GSMC	2.69 (0.20)	3.03 (0.23)	3.30 (0.05)	3.78 (0.17)	4.12 (0.02)	3.87 (0.23)	3.63 (0.18)	3.03 (0.23)	0.74 (0.36)	-6.10 (0.79)	-47.55 (0.00)
Kddcup99	99.74 (0.00)	99.75 (0.01)	99.75 (0.01)	99.74 (0.00)	99.74 (0.00)	99.74 (0.00)	99.74 (0.00)	99.74 (0.00)	99.73 (0.01)	99.73 (0.01)	99.73 (0.00)
Keystroke	88.37 (0.41)	87.88 (0.43)	87.87 (0.60)	86.72 (0.00)	87.80 (0.14)	84.99 (0.81)	85.85 (0.20)	83.37 (0.00)	82.04 (0.81)	81.17 (0.34)	76.94 (0.00)
NOAA	33.83 (0.00)	34.77 (0.35)	34.79 (0.27)	35.04 (0.03)	34.96 (0.04)	35.04 (1.00)	34.02 (0.79)	32.96 (0.02)	31.13 (0.21)	25.26 (0.40)	16.91 (0.00)
Nomao	90.83 (0.00)	90.63 (0.11)	90.58 (0.23)	90.45 (0.02)	90.39 (0.18)	90.16 (0.05)	90.29 (0.05)	90.08 (0.09)	89.98 (0.06)	90.09 (0.04)	89.33 (0.00)
Outdoor	67.57 (0.33)	67.73 (0.33)	67.85 (0.27)	68.16 (0.91)	67.76 (0.31)	66.58 (0.48)	67.93 (0.39)	67.12 (0.13)	67.26 (0.43)	67.76 (0.38)	69.46 (0.00)
Ozone	-0.62 (1.08)	-0.25 (1.70)	2.19 (0.32)	3.75 (0.00)	4.06 (0.31)	4.37 (1.25)	3.44 (1.57)	3.75 (1.25)	-0.31 (3.43)	-9.06 (5.32)	-33.12 (0.00)
Poker	79.46 (0.11)	79.56 (0.08)	79.82 (0.20)	80.04 (0.13)	80.47 (0.15)	80.68 (0.26)	80.70 (0.63)	81.29 (0.16)	82.50 (0.32)	83.26 (0.29)	89.36 (0.00)
Rialto	77.90 (0.12)	77.50 (0.06)	76.94 (0.09)	76.67 (0.10)	76.23 (0.02)	75.65 (0.12)	75.01 (0.05)	74.63 (0.17)	73.82 (0.07)	73.58 (0.12)	74.84 (0.00)
Shuttle	98.80 (0.04)	98.76 (0.02)	98.69 (0.00)	98.71 (0.05)	98.73 (0.08)	98.69 (0.02)	98.65 (0.06)	98.70 (0.01)	98.65 (0.05)	98.49 (0.09)	98.29 (0.00)
AGR	83.75 (0.09)	83.63 (0.08)	83.21 (0.15)	83.46 (0.02)	83.13 (0.11)	82.85 (0.09)	82.46 (0.03)	81.98 (0.10)	81.43 (0.00)	80.37 (0.12)	77.99 (0.00)
AGR_a	84.76 (0.23)	85.04 (0.17)	85.31 (0.46)	84.43 (0.49)	84.79 (0.40)	84.83 (0.04)	84.95 (0.02)	84.33 (0.19)	84.08 (0.13)	83.29 (0.00)	79.80 (0.00)
AGR_g	78.00 (0.29)	78.64 (0.52)	78.52 (0.59)	78.30 (0.30)	78.28 (0.29)	78.08 (0.15)	78.29 (0.13)	77.76 (0.01)	77.11 (0.23)	75.07 (0.05)	71.80 (0.00)
HYPER	70.94 (0.06)	71.27 (0.21)	71.50 (0.02)	71.77 (0.02)	72.11 (0.09)	72.36 (0.03)	72.15 (0.13)	71.88 (0.17)	70.62 (0.10)	67.43 (0.08)	58.71 (0.00)
LED	71.04 (0.00)	71.02 (0.02)	71.01 (0.03)	71.01 (0.01)	71.03 (0.02)	71.02 (0.01)	70.97 (0.02)	70.93 (0.01)	70.37 (0.02)	68.53 (0.04)	65.58 (0.00)
LED_a	71.09 (0.00)	71.10 (0.02)	71.10 (0.03)	71.09 (0.01)	71.06 (0.01)	71.02 (0.00)	71.01 (0.04)	70.87 (0.02)	70.13 (0.00)	68.14 (0.05)	65.28 (0.00)
LED_g	70.22 (0.01)	70.22 (0.01)	70.22 (0.02)	70.21 (0.02)	70.16 (0.02)	70.17 (0.02)	70.09 (0.04)	69.95 (0.04)	69.37 (0.02)	67.49 (0.05)	64.32 (0.00)
MIXED	47.08 (0.04)	50.27 (0.04)	53.96 (0.00)	57.95 (0.04)	62.43 (0.06)	67.09 (0.07)	72.15 (0.02)	77.49 (0.03)	83.08 (0.05)	89.00 (0.02)	96.53 (0.00)
RBF_f	66.24 (0.04)	66.04 (0.05)	65.88 (0.00)	65.71 (0.01)	65.39 (0.02)	65.07 (0.00)	64.74 (0.02)	64.26 (0.00)	63.50 (0.04)	62.14 (0.02)	59.56 (0.00)
RBF_m	80.03 (0.00)	79.96 (0.02)	79.89 (0.02)	79.78 (0.01)	79.74 (0.05)	79.57 (0.02)	79.43 (0.03)	79.34 (0.04)	79.12 (0.02)	78.87 (0.01)	77.12 (0.00)
RTG	81.20 (1.02)	81.28 (0.83)	81.76 (1.02)	82.67 (0.23)	84.49 (0.86)	86.12 (0.03)	85.91 (0.25)	87.24 (0.30)	87.89 (0.05)	89.15 (0.02)	87.54 (0.00)
SEA	71.34 (0.01)	71.31 (0.01)	71.29 (0.01)	71.25 (0.02)	71.17 (0.01)	71.07 (0.01)	70.86 (0.02)	70.28 (0.01)	68.04 (0.03)	59.20 (0.13)	45.36 (0.00)
SEA_a	73.65 (0.37)	74.23 (0.47)	73.42 (0.15)	73.56 (0.40)	72.95 (0.05)	71.47 (1.26)	70.71 (0.15)	66.88 (0.66)	65.13 (0.52)	63.95 (0.26)	55.02 (0.00)
SEA_g	70.46 (0.08)	70.67 (0.34)	71.06 (0.60)	70.94 (0.16)	70.97 (0.64)	69.93 (0.48)	68.14 (0.17)	67.51 (0.58)	68.99 (0.10)	64.98 (0.11)	54.46 (0.00)
SINE	99.45 (0.00)	99.44 (0.01)	99.43 (0.00)	99.41 (0.00)	99.40 (0.02)	99.39 (0.02)	99.35 (0.01)	99.36 (0.02)	99.31 (0.01)	99.22 (0.01)	98.93 (0.00)
WAVEFORM	78.69 (0.00)	78.71 (0.01)	78.68 (0.01)	78.71 (0.00)	78.70 (0.04)	78.56 (0.02)	78.44 (0.00)	78.11 (0.04)	77.36 (0.02)	75.91 (0.00)	71.47 (0.00)
WAVEFORM_g	78.18 (0.00)	78.19 (0.03)	78.20 (0.00)	78.17 (0.01)	78.16 (0.02)	78.05 (0.02)	77.80 (0.05)	77.53 (0.03)	76.76 (0.01)	75.08 (0.15)	70.60 (0.00)
Ranking médio geral	4,23	3,71	3,97	4,34	4,57	5,69	6,09	7,03	7,69	8,74	9,40

Table 8: Comparison of the accuracy metric between the ARE algorithm and the PBIS strategy applied at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARE	δ									Inc.
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	
Airlines	64,35(0,70)	63,91 (0,04)	63,67 (0,02)	63,41 (0,04)	63,01 (0,05)	62,65 (0,04)	62,25 (0,02)	61,62 (0,08)	61,26 (0,03)	60,64 (0,07)	60,02 (0,01)
BankFraud	98,89(0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,89 (0,00)	98,87 (0,00)	98,83 (0,00)	98,68 (0,01)	96,99 (0,00)
Census	95,77(0,01)	95,76 (0,00)	95,77 (0,00)	95,74 (0,00)	95,7 (0,00)	95,68 (0,00)	95,58 (0,00)	95,45 (0,02)	95,24 (0,01)	94,96 (0,01)	93,84 (0,01)
Coil2000	93,93(0,04)	93,92 (0,01)	93,93 (0,05)	93,83 (0,02)	93,77 (0,02)	93,45 (0,02)	93,17 (0,13)	93,02 (0,18)	90,86 (0,13)	88,12 (0,00)	84,71 (0,05)
Connect-4	79,57(0,33)	80,09 (0,05)	80,12 (0,06)	80,29 (0,01)	80,45 (0,10)	80,26 (0,05)	80,26 (0,01)	79,86 (0,06)	79,32 (0,14)	78,11 (0,08)	75,91 (0,02)
CovType	95,37(0,12)	95,47 (0,03)	95,45 (0,04)	95,45 (0,00)	95,45 (0,02)	95,4 (0,00)	95,42 (0,02)	95,37 (0,03)	95,27 (0,02)	95,16 (0,00)	94,36 (0,00)
Electricity	91,26(0,24)	91,28 (0,02)	91,3 (0,00)	91,37 (0,04)	91,44 (0,00)	91,43 (0,00)	91,51 (0,03)	91,59 (0,06)	91,61 (0,02)	91,47 (0,05)	90,66 (0,02)
Gassensor	92,36(0,97)	91,19 (0,00)	91,19 (0,02)	91,5 (0,45)	91,61 (0,27)	91,37 (0,10)	91,81 (0,02)	91,87 (0,32)	92,19 (0,13)	92,12 (0,08)	93,21 (0,01)
GMSC	93,44(0,02)	93,37 (0,02)	93,4 (0,01)	93,31 (0,03)	93,2 (0,02)	93,12 (0,02)	92,95 (0,00)	92,71 (0,01)	92,2 (0,01)	91,35 (0,00)	88,07 (0,02)
Kddcup99	99,88(0,00)	99,88 (0,00)	99,87 (0,00)	99,87 (0,00)	99,86 (0,01)	99,87 (0,00)	99,86 (0,01)	99,86 (0,01)	99,86 (0,01)	99,85 (0,00)	99,85 (0,00)
Keystroke	93,08(0,38)	93,50 (0,31)	92,56 (0,38)	92,75 (0,44)	92,12 (0,31)	92,18 (0,22)	91,62 (0,09)	90,93 (0,19)	91,06 (0,38)	90,43 (0,06)	87,93 (0,00)
NOAA	78,84(0,06)	78,55 (0,03)	78,53 (0,07)	77,99 (0,22)	77,98 (0,02)	77,29 (0,16)	77,13 (0,04)	76,28 (0,06)	74,79 (0,01)	73,55 (0,04)	71,71 (0,19)
Nomao	97,00(0,02)	96,98 (0,01)	96,91 (0,02)	97,04 (0,03)	96,99 (0,02)	96,99 (0,02)	97,19 (0,05)	97,35 (0,06)	97,36 (0,04)	97,38 (0,01)	96,66 (0,01)
Outdoor	70,79(2,56)	71,35 (0,15)	72,12 (0,12)	72,77 (1,23)	71,87 (1,20)	70,57 (1,25)	71,52 (0,25)	72,22 (0,23)	72,42 (0,37)	74,77 (0,22)	78,00 (0,25)
Ozone	94,03(0,11)	93,92 (0,02)	93,84 (0,04)	93,76 (0,02)	93,76 (0,06)	93,37 (0,02)	93,05 (0,08)	93,01 (0,36)	91,87 (0,11)	91,16 (0,43)	89,85 (0,00)
Poker	92,76(1,62)	90,78 (0,07)	91,03 (0,20)	91,28 (0,12)	91,17 (0,09)	91,86 (0,03)	91,74 (0,05)	91,56 (0,13)	92,14 (0,01)	92,18 (0,03)	91,21 (0,37)
Rialto	76,79(4,46)	81,63 (0,19)	81,69 (0,12)	82,11 (0,05)	82,95 (0,22)	82,59 (0,06)	82,99 (0,02)	82,93 (0,25)	82,17 (0,13)	81,64 (0,09)	80,53 (0,01)
Shuttle	99,75(0,01)	99,72 (0,00)	99,73 (0,01)	99,72 (0,00)	99,67 (0,00)	99,71 (0,02)	99,64 (0,02)	99,64 (0,01)	99,62 (0,01)	99,61 (0,02)	99,23 (0,00)
AGR	94,43(0,02)	94,37 (0,01)	94,31 (0,02)	94,27 (0,01)	94,16 (0,00)	94,03 (0,01)	93,81 (0,03)	93,57 (0,01)	92,9 (0,05)	91,83 (0,02)	90,35 (0,08)
AGR_a	87,31(2,76)	90,24 (0,10)	89,76 (0,13)	89,98 (0,04)	89,4 (0,10)	89,24 (0,23)	89,06 (0,02)	88,58 (0,02)	87,95 (0,02)	86,73 (0,22)	85,06 (0,17)
AGR_g	83,25(2,52)	85,96 (0,30)	85,59 (0,31)	85,62 (0,18)	85,8 (0,07)	85,34 (0,07)	84,8 (0,14)	84,64 (0,18)	83,36 (0,27)	82,62 (0,12)	80,6 (0,17)
HYPER	87,49(0,47)	87,11 (0,02)	87,31 (0,03)	87,38 (0,02)	87,36 (0,00)	87,17 (0,05)	86,91 (0,04)	85,67 (0,07)	83,41 (0,03)	80,56 (0,03)	76,94 (0,04)
LED	73,90(0,02)	73,90 (0,01)	73,92 (0,01)	73,90 (0,00)	73,87 (0,01)	73,65 (0,05)	73,15 (0,05)	72,48 (0,08)	71,12 (0,04)	69,17 (0,03)	66,7 (0,03)
LED_a	73,88(0,11)	73,71 (0,01)	73,63 (0,01)	73,62 (0,02)	73,67 (0,08)	73,47 (0,04)	72,98 (0,02)	72,45 (0,06)	71,39 (0,01)	69,78 (0,12)	67,40 (0,02)
LED_g	72,97(0,13)	72,78 (0,01)	72,81 (0,01)	72,75 (0,02)	72,8 (0,02)	72,55 (0,03)	72,04 (0,04)	71,46 (0,01)	70,48 (0,01)	68,76 (0,02)	66,41 (0,02)
MIXED	99,81(0,00)	99,80 (0,01)	99,80 (0,00)	99,79 (0,00)	99,78 (0,01)	99,77 (0,01)	99,76 (0,00)	99,74 (0,01)	99,71 (0,00)	99,65 (0,01)	99,35 (0,00)
RBF_f	75,69(0,47)	74,85 (0,09)	74,44 (0,15)	73,98 (0,16)	73,28 (0,09)	72,75 (0,16)	71,84 (0,13)	70,73 (0,19)	68,9 (0,14)	66,12 (0,07)	63,29 (0,08)
RBF_m	87,64(0,28)	87,15 (0,01)	87,05 (0,02)	86,97 (0,04)	86,79 (0,01)	86,65 (0,03)	86,34 (0,00)	86,03 (0,04)	85,42 (0,03)	84,2 (0,06)	81,09 (0,11)
RTG	96,75(0,09)	96,69 (0,10)	96,67 (0,11)	96,67 (0,05)	96,6 (0,07)	96,56 (0,18)	96,28 (0,02)	96,25 (0,05)	95,72 (0,13)	95,37 (0,05)	93,44 (0,06)
SEA	89,80(0,00)	89,77 (0,00)	89,75 (0,01)	89,69 (0,00)	89,58 (0,01)	89,35 (0,01)	88,73 (0,04)	87,28 (0,06)	84,98 (0,08)	81,79 (0,04)	78,23 (0,02)
SEA_a	89,58(0,00)	89,54 (0,01)	89,48 (0,01)	89,4 (0,00)	89,27 (0,00)	88,99 (0,02)	88,33 (0,02)	86,95 (0,03)	84,83 (0,06)	81,86 (0,03)	78,35 (0,02)
SEA_g	89,15(0,01)	89,1 (0,01)	89,06 (0,00)	88,98 (0,00)	88,8 (0,01)	88,52 (0,01)	87,89 (0,01)	86,59 (0,05)	84,5 (0,10)	81,44 (0,03)	78,05 (0,04)
SINE	99,86(0,00)	99,86 (0,00)	99,85 (0,00)	99,85 (0,00)	99,84 (0,01)	99,84 (0,01)	99,82 (0,00)	99,79 (0,00)	99,76 (0,00)	99,68 (0,00)	99,14 (0,01)
WAVEFORM	85,80(0,03)	85,76 (0,01)	85,67 (0,01)	85,56 (0,01)	85,47 (0,01)	85,24 (0,00)	84,92 (0,02)	84,49 (0,02)	83,74 (0,03)	82,6 (0,00)	79,81 (0,01)
WAVEFORM_g	85,57(0,02)	85,46 (0,02)	85,37 (0,02)	85,23 (0,03)	85,12 (0,01)	84,9 (0,02)	84,54 (0,05)	84,03 (0,06)	83,31 (0,04)	82,03 (0,01)	79,16 (0,02)
Ranking médio geral	2,94	3,54	3,77	3,89	4,66	5,66	6,14	7,14	7,80	8,89	10,29

Table 9: Comparison of the Kappa M metric between the ARE algorithm and the application of the PBIS strategy at the *ensemble* level. Bold values represent the best results per metric.

Dataset	ARE	δ									Inc.
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	
Airlines	19,98(1,58)	18,98(0,08)	18,44(0,05)	17,85(0,09)	16,96(0,12)	16,15(0,09)	15,25(0,05)	13,84(0,17)	13,03(0,07)	11,64(0,15)	10,26(0,03)
BankFraud	0,00(0,00)	0,02(0,03)	-0,02(0,01)	0,00(0,00)	0,02(0,09)	-0,17(0,01)	-0,56(0,14)	-1,71(0,06)	-5,50(0,05)	-19,00(0,92)	-172,07(0,42)
Census	31,91(0,12)	31,66(0,11)	31,83(0,01)	31,35(0,06)	30,73(0,05)	30,42(0,02)	28,90(0,01)	26,75(0,28)	23,34(0,05)	18,82(0,25)	0,78(0,15)
Coil2000	-1,70(0,61)	-1,87(0,17)	-1,70(0,86)	-3,41(0,17)	-4,26(0,34)	-9,72(0,43)	-14,33(2,05)	-16,89(3,07)	-53,07(2,14)	-98,97(0,09)	-156,14(0,94)
Connect-4	40,23(0,98)	41,75(0,13)	41,82(0,20)	42,34(0,04)	42,80(0,29)	42,23(0,16)	42,23(0,03)	41,06(0,19)	39,49(0,41)	35,94(0,22)	29,51(0,04)
CovType	90,96(0,24)	91,16(0,05)	91,13(0,06)	91,12(0,00)	91,12(0,03)	91,03(0,01)	91,07(0,04)	90,97(0,05)	90,78(0,03)	90,56(0,00)	88,99(0,01)
Electricity	79,41(0,57)	79,45(0,05)	79,51(0,01)	79,69(0,10)	79,84(0,01)	79,81(0,00)	80,01(0,06)	80,20(0,13)	80,25(0,05)	79,92(0,13)	77,99(0,04)
Gassensor	90,03(1,26)	88,49(0,00)	88,49(0,02)	88,90(0,59)	89,05(0,35)	88,73(0,13)	89,31(0,02)	89,39(0,42)	89,81(0,16)	89,71(0,09)	91,13(0,02)
GMSC	1,90(0,21)	0,92(0,34)	1,27(0,13)	0,00(0,44)	-1,68(0,21)	-2,94(0,30)	-5,42(0,02)	-8,98(0,06)	-16,63(0,16)	-29,34(0,07)	-78,36(0,30)
Kddcup99	99,72(0,01)	99,72(0,00)	99,71(0,00)	99,71(0,00)	99,67(0,02)	99,71(0,00)	99,69(0,01)	99,68(0,01)	99,69(0,02)	99,67(0,01)	99,66(0,00)
Keystroke	85,17(0,81)	86,05(0,67)	84,04(0,81)	84,45(0,93)	83,10(0,68)	83,24(0,47)	82,03(0,20)	80,56(0,40)	80,83(0,81)	79,49(0,13)	74,12(0,00)
NOAA	32,59(0,20)	31,66(0,11)	31,59(0,23)	29,88(0,69)	29,85(0,08)	27,64(0,53)	27,13(0,14)	24,41(0,20)	19,67(0,04)	15,72(0,13)	9,84(0,60)
Nomao	89,52(0,08)	89,43(0,02)	89,21(0,05)	89,64(0,09)	89,46(0,04)	89,48(0,09)	90,17(0,17)	90,72(0,20)	90,77(0,13)	90,83(0,03)	88,31(0,02)
Outdoor	69,96(2,63)	70,53(0,16)	71,33(0,12)	72,00(1,26)	71,07(1,23)	69,74(1,29)	70,71(0,25)	71,43(0,24)	71,64(0,38)	74,06(0,21)	77,37(0,26)
Ozone	5,50(1,74)	3,75(0,31)	2,50(0,63)	1,25(0,32)	1,25(0,94)	-5,00(0,32)	-9,99(1,25)	-10,62(5,63)	-28,74(1,88)	-39,99(6,88)	-60,62(0,00)
Poker	85,50(3,25)	81,53(0,15)	82,02(0,38)	82,53(0,24)	82,31(0,17)	83,69(0,05)	83,44(0,12)	83,08(0,27)	84,26(0,03)	84,33(0,06)	82,39(0,74)
Rialto	74,21(4,95)	79,59(0,21)	79,66(0,13)	80,12(0,05)	81,05(0,24)	80,66(0,07)	81,10(0,02)	81,03(0,27)	80,18(0,14)	79,61(0,10)	78,36(0,01)
Shuttle	98,83(0,04)	98,72(0,02)	98,75(0,02)	98,69(0,00)	98,47(0,01)	98,65(0,09)	98,35(0,08)	98,34(0,05)	98,22(0,03)	98,21(0,06)	96,44(0,00)
AGR	83,02(0,05)	82,85(0,03)	82,66(0,08)	82,53(0,04)	82,22(0,01)	81,79(0,04)	81,13(0,09)	80,40(0,02)	78,36(0,17)	75,10(0,08)	70,58(0,24)
AGR_a	73,10(5,84)	79,32(39,66)	78,29(0,27)	78,77(0,08)	77,52(0,22)	77,20(0,48)	76,81(0,05)	75,79(0,05)	74,45(0,05)	71,88(0,46)	68,34(0,35)
AGR_g	64,51(5,35)	70,24(35,12)	69,45(0,66)	69,53(0,38)	69,91(0,14)	68,93(0,15)	67,78(0,30)	67,43(0,38)	64,73(0,56)	63,17(0,25)	58,89(0,35)
HYPER	74,94(0,95)	74,18(0,04)	74,58(0,06)	74,73(0,05)	74,69(0,01)	74,30(0,11)	73,78(0,09)	71,30(0,15)	66,76(0,06)	61,05(0,07)	53,80(0,09)
LED	70,98(0,03)	70,98(0,01)	71,00(0,01)	70,98(0,01)	70,94(0,00)	70,70(0,05)	70,14(0,05)	69,39(0,09)	67,89(0,05)	65,72(0,03)	62,97(0,04)
LED_a	70,93(0,12)	70,74(35,37)	70,65(0,01)	70,64(0,02)	70,70(0,09)	70,48(0,05)	69,93(0,01)	69,34(0,07)	68,16(0,00)	66,36(0,13)	63,72(0,03)
LED_g	69,92(0,15)	69,70(34,85)	69,74(0,01)	69,67(0,02)	69,72(0,02)	69,44(0,03)	68,88(0,05)	68,23(0,01)	67,14(0,01)	65,23(0,02)	62,61(0,02)
MIXED	99,38(0,01)	99,36(0,01)	99,33(0,01)	99,32(0,01)	99,29(0,01)	99,26(0,01)	99,21(0,01)	99,16(0,01)	99,05(0,00)	98,83(0,01)	97,84(0,00)
RBF_f	65,27(0,67)	64,07(32,04)	63,48(0,21)	62,83(0,23)	61,82(0,13)	61,07(0,23)	59,77(0,19)	58,18(0,27)	55,56(0,20)	51,59(0,11)	47,55(0,12)
RBF_m	82,34(0,40)	81,64(40,82)	81,50(0,03)	81,38(0,05)	81,12(0,02)	80,92(0,03)	80,48(0,00)	80,04(0,06)	79,17(0,05)	77,43(0,08)	72,98(0,16)
RTG	92,29(0,20)	92,16(46,08)	92,10(0,24)	92,10(0,13)	91,95(0,16)	91,84(0,42)	91,19(0,05)	91,11(0,12)	89,85(0,31)	89,02(0,10)	84,44(0,14)
SEA	71,41(0,01)	71,33(0,00)	71,27(0,01)	71,10(0,01)	70,80(0,02)	70,15(0,02)	68,41(0,10)	64,36(0,17)	57,89(0,21)	48,98(0,13)	38,98(0,03)
SEA_a	75,85(0,01)	75,76(37,88)	75,63(0,02)	75,43(0,00)	75,14(0,00)	74,48(0,03)	72,95(0,03)	69,76(0,07)	64,84(0,14)	57,96(0,07)	49,82(0,02)
SEA_g	74,87(0,02)	74,74(37,37)	74,65(0,00)	74,45(0,01)	74,05(0,01)	73,39(0,02)	71,95(0,04)	68,93(0,11)	64,07(0,22)	56,99(0,08)	49,12(0,08)
SINE	99,70(0,00)	99,71(0,00)	99,68(0,00)	99,67(0,01)	99,65(0,01)	99,65(0,01)	99,61(0,00)	99,55(0,00)	99,49(0,00)	99,31(0,00)	98,13(0,02)
WAVEFORM	78,66(0,05)	78,60(39,30)	78,46(0,02)	78,29(0,02)	78,16(0,02)	77,82(0,01)	77,34(0,02)	76,69(0,02)	75,56(0,05)	73,86(0,02)	69,65(0,01)
WAVEFORM_g	78,32(0,03)	78,16(39,08)	78,01(0,03)	77,81(0,05)	77,64(0,02)	77,32(0,02)	76,77(0,08)	76,00(0,09)	74,93(0,05)	73,01(0,01)	68,69(0,04)
Ranking médio geral	2,97	3,51	3,91	4,03	4,80	5,80	6,31	7,23	7,80	8,86	10,31

Table 10: Comparison of the accuracy metric between the ARE algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	64,35(0,70)	63,94(0,20)	63,70(0,28)	63,42(0,28)	63,12(0,27)	62,82(0,26)	62,41(0,29)	62,01(0,28)	61,56(0,23)	60,96(0,21)	60,43(0,18)
BankFraud	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,88(0,01)	98,87(0,00)	98,61(0,06)
Census	95,77(0,01)	95,79(0,04)	95,79(0,02)	95,81(0,03)	95,84(0,04)	95,84(0,02)	95,86(0,04)	95,77(0,02)	95,74(0,02)	95,77(0,06)	95,47(0,02)
Coil2000	93,93(0,04)	93,94(0,02)	93,90(0,02)	93,87(0,03)	93,89(0,01)	93,77(0,02)	93,61(0,06)	93,30(0,04)	92,18(0,27)	90,40(0,28)	87,23(0,81)
Connect-4	79,57(0,33)	80,30(0,03)	80,37(0,02)	80,52(0,04)	80,47(0,05)	80,56(0,10)	80,62(0,13)	80,75(0,01)	80,68(0,14)	80,67(0,04)	80,32(0,15)
CovType	95,37(0,12)	95,48(0,11)	95,53(0,13)	95,56(0,13)	95,62(0,15)	95,69(0,13)	95,75(0,13)	95,83(0,12)	95,92(0,10)	96,01(0,10)	96,09(0,12)
Electricity	91,26(0,24)	91,28(0,13)	91,41(0,16)	91,50(0,22)	91,64(0,24)	91,65(0,08)	91,82(0,15)	91,78(0,04)	92,03(0,09)	92,20(0,11)	92,54(0,17)
Gassensor	92,36(0,97)	91,46(0,69)	91,59(0,87)	91,81(1,08)	91,92(0,93)	92,09(0,84)	92,38(0,93)	92,60(0,73)	93,59(1,23)	93,64(0,88)	94,18(1,12)
GMSC	93,44(0,02)	93,41(0,04)	93,40(0,03)	93,39(0,01)	93,32(0,05)	93,30(0,05)	93,23(0,05)	93,17(0,06)	93,05(0,07)	92,89(0,06)	91,84(0,34)
Kddcup99	99,88(0,00)	99,87(0,00)	99,87(0,00)	99,88(0,00)	99,88(0,00)	99,87(0,00)	99,87(0,01)	99,87(0,02)	99,86(0,02)	99,87(0,02)	99,87(0,02)
Keystroke	93,08(0,38)	93,18(0,03)	93,06(0,06)	93,00(0,19)	92,75(0,22)	93,50(0,13)	93,18(0,03)	92,50(0,16)	92,68(0,53)	92,75(0,47)	92,31(0,16)
NOAA	78,84(0,06)	78,50(0,17)	78,45(0,06)	78,26(0,14)	78,20(0,20)	78,38(0,05)	78,14(0,15)	77,76(0,13)	77,76(0,12)	77,10(0,16)	76,82(0,29)
Nomao	97,00(0,02)	96,92(0,09)	96,93(0,11)	96,98(0,06)	96,97(0,13)	97,04(0,13)	97,10(0,08)	97,13(0,12)	97,49(0,03)	97,61(0,03)	97,46(0,10)
Outdoor	70,79(2,56)	73,42(0,16)	72,25(0,16)	71,25(0,02)	72,15(0,17)	71,52(0,32)	72,85(0,42)	71,87(0,20)	71,67(0,80)	74,85(1,03)	74,25(0,11)
Ozone	94,03(0,11)	93,80(0,14)	93,88(0,04)	94,04(0,06)	93,72(0,10)	93,29(0,23)	93,25(0,12)	93,01(0,08)	92,46(0,36)	92,22(0,08)	91,83(0,18)
Poker	92,76(1,62)	90,98(0,80)	91,60(1,18)	91,85(1,33)	92,55(1,40)	92,51(1,26)	93,49(1,34)	93,63(1,09)	94,37(1,16)	94,27(0,79)	94,72(1,14)
Rialto	76,79(4,46)	81,66(0,14)	82,04(0,17)	82,94(0,09)	83,05(0,13)	83,13(0,02)	83,86(0,18)	83,77(0,02)	84,16(0,21)	84,56(0,26)	84,90(0,48)
Shuttle	99,75(0,01)	99,72(0,00)	99,72(0,01)	99,71(0,01)	99,69(0,00)	99,71(0,00)	99,73(0,01)	99,72(0,01)	99,70(0,01)	99,70(0,00)	99,71(0,00)
AGR	94,43(0,02)	94,42(0,04)	94,40(0,06)	94,40(0,04)	94,34(0,07)	94,29(0,08)	94,28(0,06)	94,15(0,09)	94,02(0,06)	93,60(0,13)	92,79(0,24)
AGR_a	87,31(2,76)	89,96(0,19)	90,00(0,40)	89,99(0,38)	89,41(0,07)	89,33(0,19)	88,83(0,02)	88,80(0,30)	88,24(0,31)	87,53(0,20)	88,76(0,02)
AGR_g	83,25(2,52)	85,45(0,12)	85,93(0,28)	85,55(0,06)	85,49(0,16)	85,37(0,19)	85,32(0,28)	85,07(0,01)	84,85(0,10)	85,01(0,07)	85,08(0,25)
HYPER	87,49(0,47)	87,03(0,05)	87,11(0,05)	87,05(0,12)	87,09(0,10)	87,04(0,11)	87,05(0,09)	86,60(0,00)	86,27(0,05)	85,75(0,19)	85,54(0,21)
LED	73,90(0,02)	73,94(0,02)	73,89(0,02)	73,88(0,03)	73,82(0,04)	73,70(0,01)	73,57(0,01)	73,38(0,12)	73,23(0,23)	73,07(0,27)	72,72(0,48)
LED_a	73,88(0,11)	73,54(0,06)	73,72(0,09)	73,65(0,13)	73,65(0,20)	73,55(0,25)	73,39(0,10)	73,36(0,09)	73,15(0,05)	73,02(0,02)	72,95(0,02)
LED_g	72,97(0,13)	72,82(0,02)	72,82(0,02)	72,71(0,11)	72,56(0,12)	72,49(0,13)	72,59(0,19)	72,37(0,05)	72,31(0,13)	72,27(0,04)	72,16(0,01)
MIXED	99,81(0,00)	99,79(0,00)	99,79(0,00)	99,79(0,00)	99,79(0,00)	99,79(0,00)	99,79(0,01)	99,78(0,00)	99,79(0,00)	99,78(0,01)	99,78(0,00)
RBF_f	75,69(0,47)	75,25(0,17)	75,20(0,27)	75,27(0,38)	75,28(0,50)	75,31(0,54)	75,10(0,57)	75,08(0,66)	75,00(0,62)	74,98(0,73)	74,88(0,74)
RBF_lm	87,64(0,28)	87,25(0,05)	87,24(0,06)	87,21(0,10)	87,22(0,13)	87,22(0,13)	87,23(0,15)	87,26(0,18)	87,25(0,16)	87,30(0,17)	87,29(0,19)
RTG	96,75(0,09)	96,62(0,08)	96,67(0,25)	96,76(0,33)	96,80(0,35)	96,75(0,22)	96,71(0,20)	96,90(0,23)	96,93(0,24)	96,89(0,16)	96,88(0,16)
SEA	89,80(0,00)	89,77(0,02)	89,75(0,02)	89,74(0,03)	89,70(0,03)	89,65(0,03)	89,52(0,00)	89,24(0,10)	88,82(0,22)	88,18(0,41)	87,24(0,65)
SEA_a	89,58(0,00)	89,55(0,00)	89,51(0,00)	89,48(0,01)	89,43(0,01)	89,38(0,02)	89,18(0,04)	88,83(0,15)	88,32(0,28)	87,66(0,47)	87,00(0,59)
SEA_g	89,15(0,01)	89,13(0,02)	89,09(0,02)	89,07(0,04)	89,01(0,05)	88,91(0,04)	88,75(0,02)	88,46(0,07)	88,07(0,18)	87,40(0,40)	86,59(0,61)
SINE	99,86(0,00)	99,86(0,01)	99,86(0,02)	99,85(0,01)	99,85(0,01)	99,85(0,02)	99,85(0,02)	99,85(0,02)	99,84(0,02)	99,83(0,02)	99,80(0,02)
WAVEFORM	85,80(0,03)	85,82(0,01)	85,77(0,02)	85,75(0,02)	85,75(0,01)	85,68(0,03)	85,61(0,05)	85,62(0,02)	85,53(0,04)	85,46(0,07)	85,38(0,05)
WAVEFORM_g	85,57(0,02)	85,51(0,00)	85,50(0,02)	85,46(0,04)	85,41(0,05)	85,40(0,03)	85,34(0,07)	85,33(0,04)	85,24(0,05)	85,17(0,06)	85,14(0,06)
Ranking médio geral	4,09	4,60	4,54	4,91	5,17	5,40	5,40	6,31	7,23	7,34	8,00

Table 11: Comparison of the Kappa M metric between the ARE algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	19,98(1,58)	19.05(0.46)	18.52(0.63)	17.89(0.62)	17.22(0.61)	16.53(0.59)	15.62(0.66)	14.71(0.63)	13.71(0.52)	12.36(0.47)	11.17(0.41)
BankFraud	0,00(0,00)	-0.01(0.01)	0.01(0.00)	0.00(0.00)	-0.03(0.02)	-0.01(0.00)	-0.04(0.02)	-0.31(0.13)	-0.79(0.16)	-2.01(0.38)	-25.68(6.00)
Census	31,91(0,12)	32.20(0.66)	32.20(0.34)	32.55(0.47)	33.05(0.55)	33.08(0.32)	33.28(0.59)	31.83(0.27)	31.37(0.27)	31.82(0.96)	27.05(0.33)
Coil2000	-1,70(0,61)	-1.53(0.34)	-2.21(0.34)	-2.73(0.52)	-2.38(0.09)	-4.26(0.43)	-6.99(1.03)	-12.28(0.60)	-31.05(4.52)	-60.75(4.70)	-113.99(13.48)
Connect-4	40,23(0,98)	42.37(0.11)	42.57(0.05)	43.00(0.12)	42.86(0.15)	43.13(0.29)	43.28(0.39)	43.67(0.04)	43.47(0.41)	43.44(0.11)	42.41(0.45)
CovType	90,96(0,24)	91.18(0.20)	91.28(0.24)	91.34(0.26)	91.46(0.29)	91.60(0.25)	91.72(0.27)	91.87(0.23)	92.05(0.20)	92.21(0.18)	92.38(0.23)
Electricity	79,41(0,57)	79.46(0.32)	79.78(0.38)	79.99(0.52)	80.31(0.56)	80.33(0.19)	80.74(0.36)	80.65(0.10)	81.24(0.23)	81.63(0.26)	82.43(0.40)
Gassensor	90,03(1,26)	88.85(0.90)	89.02(1.13)	89.31(1.41)	89.45(1.21)	89.68(1.11)	90.05(1.22)	90.33(0.95)	91.63(1.61)	91.70(1.15)	92.40(1.47)
GMSC	1,90(0,21)	1.47(0.59)	1.28(0.55)	1.14(0.17)	0.07(0.71)	-0.16(0.81)	-1.27(0.83)	-2.06(0.97)	-3.94(1.05)	-6.37(0.97)	-21.95(5.06)
Kddcup99	99,72(0,01)	99.71(0.00)	99.71(0.01)	99.72(0.01)	99.73(0.02)	99.71(0.00)	99.70(0.01)	99.71(0.03)	99.68(0.06)	99.71(0.03)	99.72(0.05)
Keystroke	85,17(0,81)	85.38(0.07)	85.12(0.14)	84.98(0.41)	84.45(0.46)	86.05(0.27)	85.38(0.06)	83.91(0.34)	84.31(1.14)	84.45(1.01)	83.51(0.34)
NOAA	32,59(0,20)	31.50(0.55)	31.34(0.20)	30.73(0.47)	30.53(0.63)	31.11(0.15)	30.34(0.48)	29.13(0.40)	29.15(0.38)	27.04(0.51)	26.14(0.92)
Nomao	89,52(0,08)	89.24(0.32)	89.26(0.38)	89.43(0.21)	89.42(0.44)	89.65(0.44)	89.87(0.26)	89.97(0.42)	91.22(0.13)	91.65(0.10)	91.14(0.33)
Outdoor	69,96(2,63)	72.67(0.17)	71.46(0.16)	70.43(0.01)	71.36(0.17)	70.71(0.33)	72.08(0.44)	71.07(0.21)	70.87(0.82)	74.13(1.05)	73.52(0.12)
Ozone	5,50(1,74)	1.87(2.19)	3.12(0.63)	5.62(0.93)	0.62(1.57)	-6.24(3.75)	-6.87(1.88)	-10.62(1.25)	-19.37(5.63)	-23.12(1.25)	-29.37(2.81)
Poker	85,50(3,25)	81.92(1.61)	83.16(2.36)	83.67(2.67)	85.08(2.81)	84.99(2.52)	86.95(2.67)	87.23(2.18)	88.72(2.32)	88.51(1.58)	89.41(2.28)
Rialto	74,21(4,95)	79.62(0.16)	80.05(0.19)	81.05(0.10)	81.17(0.14)	81.26(0.02)	82.07(0.20)	81.96(0.04)	82.40(0.23)	82.84(0.29)	83.22(0.52)
Shuttle	98,83(0,04)	98.70(0.01)	98.72(0.05)	98.68(0.05)	98.59(0.00)	98.65(0.01)	98.74(0.00)	98.69(0.02)	98.63(0.02)	98.60(0.03)	98.67(0.00)
AGR	83,02(0,05)	82.99(0.12)	82.92(0.19)	82.94(0.13)	82.74(0.23)	82.61(0.23)	82.58(0.17)	82.17(0.30)	81.77(0.19)	80.50(0.39)	78.04(0.72)
AGR_a	73,10(5,84)	78.73(0.41)	78.80(0.84)	78.78(0.80)	77.56(0.15)	77.38(0.40)	76.32(0.06)	76.27(0.63)	75.07(0.64)	73.57(0.42)	76.17(0.05)
AGR_g	64,51(5,35)	69.16(0.25)	70.19(0.59)	69.36(0.13)	69.25(0.34)	69.00(0.41)	68.89(0.60)	68.35(0.03)	67.88(0.21)	68.23(0.15)	68.38(0.53)
HYPER	74,94(0,95)	74.02(0.09)	74.18(0.09)	74.06(0.23)	74.14(0.21)	74.04(0.23)	74.07(0.18)	73.16(0.01)	72.49(0.09)	71.46(0.38)	71.04(0.43)
LED	70,98(0,03)	71.02(0.03)	70.96(0.02)	70.95(0.04)	70.89(0.05)	70.75(0.01)	70.61(0.02)	70.40(0.13)	70.23(0.25)	70.05(0.30)	69.66(0.53)
LED_a	70,93(0,12)	70.55(0.07)	70.76(0.11)	70.67(0.14)	70.68(0.23)	70.57(0.29)	70.39(0.11)	70.35(0.10)	70.12(0.05)	69.97(0.02)	69.89(0.01)
LED_g	69,92(0,15)	69.75(0.02)	69.75(0.02)	69.62(0.12)	69.46(0.13)	69.38(0.14)	69.49(0.20)	69.25(0.05)	69.18(0.14)	69.14(0.05)	69.01(0.01)
MIXED	99,38(0,01)	99.32(0.00)	99.31(0.01)	99.30(0.00)	99.30(0.01)	99.30(0.01)	99.30(0.00)	99.29(0.01)	99.30(0.00)	99.28(0.01)	99.27(0.00)
RBF_f	65,27(0,67)	64.64(0.24)	64.57(0.39)	64.67(0.54)	64.68(0.71)	64.72(0.77)	64.42(0.81)	64.40(0.94)	64.28(0.89)	64.25(1.04)	64.11(1.06)
RBF_lm	82,34(0,40)	81.79(0.08)	81.77(0.09)	81.72(0.14)	81.74(0.18)	81.75(0.19)	81.76(0.21)	81.79(0.25)	81.78(0.22)	81.86(0.26)	81.84(0.27)
RTG	92,29(0,20)	91.99(0.17)	92.12(0.60)	92.31(0.77)	92.42(0.83)	92.30(0.52)	92.20(0.46)	92.64(0.53)	92.72(0.56)	92.62(0.37)	92.61(0.40)
SEA	71,41(0,01)	71.33(0.07)	71.28(0.07)	71.24(0.09)	71.14(0.09)	70.99(0.09)	70.62(0.01)	69.83(0.28)	68.67(0.62)	66.86(1.16)	64.23(1.83)
SEA_a	75,85(0,01)	75.78(0.00)	75.69(0.01)	75.62(0.01)	75.51(0.02)	75.38(0.04)	74.92(0.09)	74.11(0.34)	72.93(0.65)	71.41(1.09)	69.89(1.36)
SEA_g	74,87(0,02)	74.82(0.06)	74.71(0.05)	74.67(0.09)	74.53(0.11)	74.30(0.09)	73.92(0.03)	73.26(0.14)	72.34(0.41)	70.81(0.92)	68.93(1.42)
SINE	99,70(0,00)	99.71(0.03)	99.70(0.02)	99.69(0.02)	99.69(0.02)	99.68(0.03)	99.67(0.02)	99.67(0.04)	99.65(0.04)	99.63(0.04)	99.57(0.05)
WAVEFORM	78,66(0,05)	78.70(0.00)	78.62(0.03)	78.58(0.05)	78.58(0.02)	78.48(0.05)	78.38(0.08)	78.38(0.05)	78.26(0.05)	78.15(0.09)	78.03(0.06)
WAVEFORM_g	78,32(0,03)	78.23(0.00)	78.21(0.04)	78.16(0.06)	78.07(0.09)	78.07(0.05)	77.97(0.11)	77.95(0.05)	77.82(0.09)	77.73(0.09)	77.67(0.10)
Ranking médio geral	4,11	4,69	4,66	5,14	5,43	5,66	5,86	6,71	7,31	7,46	8,06

Table 12: Comparison of the accuracy metric between the ARF algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARF	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	66,65(0,02)	66,60(0,00)	66,51(0,02)	66,39(0,02)	66,22(0,00)	65,97(0,02)	65,66(0,01)	65,19(0,01)	64,49(0,01)	63,36(0,01)	61,22(0,00)
BankFraud	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,89(0,00)	98,84(0,00)
Census	95,30 (0,01)	95,32(0,01)	95,36(0,02)	95,39(0,00)	95,43(0,01)	95,47(0,00)	95,52(0,02)	95,59(0,01)	95,66(0,00)	95,71(0,01)	95,32(0,00)
Coil2000	93,90 (0,00)	93,90(0,01)	93,92(0,00)	93,93(0,00)	93,91(0,02)	93,92(0,02)	93,91(0,02)	93,90(0,04)	93,85(0,01)	93,72(0,01)	88,58(0,00)
Connect-4	78,01 (0,05)	78,10(0,04)	78,18(0,05)	78,28(0,07)	78,32(0,04)	78,43(0,06)	78,62(0,05)	78,87(0,01)	79,40(0,02)	80,37(0,06)	79,84(0,00)
CovType	94,71 (0,00)	94,73(0,02)	94,75(0,01)	94,79(0,02)	94,84(0,02)	94,90(0,02)	94,97(0,02)	95,04(0,02)	95,18(0,03)	95,44(0,01)	95,85(0,00)
Electricity	90,59 (0,06)	90,64(0,02)	90,64(0,08)	90,70(0,07)	90,74(0,05)	90,75(0,04)	90,76(0,01)	90,85(0,02)	90,96(0,03)	91,00(0,08)	91,59(0,00)
Gassensor	90,47 (0,10)	89,77(0,11)	89,90(0,07)	90,07(0,08)	89,59(0,13)	89,39(0,15)	88,34(0,32)	87,36(0,10)	88,11(0,18)	89,06(0,17)	90,60(0,00)
GMSC	93,57(0,02)	93,56(0,00)	93,57(0,02)	93,56(0,00)	93,56(0,02)	93,56(0,01)	93,54(0,02)	93,52(0,02)	93,47(0,01)	93,35(0,01)	92,71(0,00)
Kddcup99	99,86(0,00)	99,89(0,00)	99,88(0,00)	99,88(0,00)	99,88(0,00)	99,88(0,00)	99,86(0,00)	99,87(0,00)	99,84(0,00)	99,81(0,01)	99,53(0,00)
Keystroke	91,92 (0,17)	92,35(0,25)	92,18(0,18)	92,31(0,27)	92,39(0,30)	92,69(0,49)	92,37(0,13)	92,50(0,15)	92,39(0,41)	92,18(0,35)	92,25(0,00)
NOAA	79,30(0,03)	79,31(0,06)	79,37(0,05)	79,23(0,11)	79,10(0,17)	79,13(0,13)	79,11(0,02)	78,99(0,22)	78,90(0,17)	78,45(0,08)	77,33(0,00)
Nomao	97,24 (0,05)	97,21(0,02)	97,19(0,04)	97,25(0,01)	97,22(0,02)	97,16(0,03)	97,17(0,01)	97,21(0,02)	97,17(0,05)	97,28(0,04)	97,64(0,00)
Outdoor	63,79 (0,54)	65,25(0,28)	63,86(0,27)	63,52(0,10)	63,45(0,13)	64,08(0,03)	64,88(0,10)	65,38(0,05)	66,12(0,16)	67,05(0,14)	70,22(0,00)
Ozone	93,67 (0,06)	93,72(0,06)	93,72(0,03)	93,77(0,05)	93,92(0,06)	93,88(0,03)	94,04(0,03)	94,01(0,08)	94,05(0,10)	93,65(0,11)	91,98(0,00)
Poker	88,78 (0,02)	88,94(0,03)	89,12(0,03)	89,30(0,03)	89,55(0,03)	89,81(0,02)	90,25(0,02)	90,75(0,04)	91,36(0,06)	92,15(0,07)	92,36(0,00)
Rialto	72,21(0,02)	72,11(0,06)	71,97(0,15)	71,64(0,04)	71,44(0,10)	71,05(0,14)	70,50(0,12)	70,43(0,18)	70,87(0,30)	71,49(0,11)	72,54(0,00)
AGR	94,07 (0,01)	94,15(0,02)	94,23(0,00)	94,28(0,00)	94,34(0,02)	94,37(0,01)	94,38(0,02)	94,40(0,02)	94,35(0,00)	94,18(0,02)	93,43(0,00)
AGR_a	87,85 (0,08)	87,84(0,06)	87,70(0,06)	87,50(0,12)	87,39(0,07)	87,10(0,03)	86,42(0,09)	85,99(0,05)	85,53(0,24)	84,50(0,05)	85,24(0,00)
AGR_g	82,40 (0,05)	82,55(0,10)	82,64(0,13)	82,52(0,06)	82,39(0,02)	82,18(0,03)	81,97(0,15)	81,62(0,13)	81,32(0,09)	81,15(0,14)	82,24(0,00)
HYPER	85,73 (0,01)	85,79(0,03)	85,83(0,02)	85,89(0,01)	86,01(0,00)	86,09(0,01)	86,22(0,03)	86,39(0,01)	86,60(0,04)	86,54(0,02)	85,24(0,00)
LED	73,90 (0,01)	73,90(0,01)	73,90(0,01)	73,89(0,00)	73,91(0,03)	73,90(0,01)	73,91(0,01)	73,91(0,02)	73,89(0,01)	73,88(0,01)	72,45(0,00)
LED_a	73,95 (0,00)	73,94(0,01)	73,95(0,01)	73,96(0,01)	73,93(0,01)	73,95(0,01)	73,92(0,00)	73,90(0,01)	73,87(0,01)	73,76(0,02)	73,37(0,00)
LED_g	73,12(0,01)	73,11(0,01)	73,10(0,01)	73,11(0,00)	73,09(0,00)	73,08(0,01)	73,08(0,02)	73,05(0,00)	73,03(0,01)	73,00(0,02)	72,47(0,00)
RBF_f	76,28 (0,00)	76,13(0,01)	75,94(0,00)	75,66(0,02)	75,37(0,00)	75,05(0,02)	74,88(0,04)	74,96(0,01)	75,19(0,01)	75,39(0,03)	75,46(0,00)
RBF_m	87,17 (0,01)	87,21(0,01)	87,26(0,02)	87,28(0,01)	87,29(0,01)	87,35(0,01)	87,36(0,01)	87,36(0,01)	87,37(0,01)	87,36(0,02)	87,39(0,00)
RTG	96,09 (0,07)	95,87(0,07)	95,69(0,05)	95,57(0,11)	95,36(0,05)	95,20(0,06)	94,98(0,12)	94,98(0,06)	95,02(0,04)	95,01(0,03)	94,51(0,00)
SEA	89,67(0,00)	89,68(0,00)	89,68(0,00)	89,66(0,01)	89,65(0,00)	89,63(0,01)	89,59(0,00)	89,55(0,00)	89,47(0,00)	89,30(0,01)	88,50(0,00)
SEA_a	89,46(0,01)	89,46(0,00)	89,44(0,01)	89,41(0,00)	89,36(0,02)	89,33(0,00)	89,29(0,01)	89,23(0,00)	89,13(0,00)	88,89(0,01)	88,01(0,00)
SEA_g	88,77(0,35)	88,97(0,01)	88,97(0,00)	88,97(0,01)	88,95(0,00)	88,93(0,01)	88,89(0,00)	88,82(0,00)	88,72(0,01)	88,49(0,01)	87,67(0,00)
SINE	99,52(0,00)	99,52(0,00)	99,52(0,01)	99,53(0,00)	99,52(0,01)	99,49(0,00)	99,49(0,00)	99,46(0,00)	99,45(0,00)	99,39(0,00)	99,21(0,00)
WAVEFORM	85,46 (0,03)	85,52(0,01)	85,54(0,00)	85,59(0,02)	85,64(0,02)	85,66(0,00)	85,68(0,00)	85,70(0,02)	85,70(0,01)	85,64(0,02)	85,46(0,00)
WAVEFORM_g	85,18 (0,00)	85,22(0,01)	85,25(0,03)	85,26(0,01)	85,32(0,01)	85,33(0,02)	85,37(0,02)	85,38(0,02)	85,37(0,01)	85,32(0,00)	85,18(0,00)
Ranking médio geral	5,85	5,21	5,06	4,79	5,79	5,42	5,76	5,73	6,18	6,88	7,67

Table 13: Comparison of the Kappa M metric between the ARF algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARF	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	25,14 (0,05)	25,03(0,01)	24,82(0,03)	24,56(0,04)	24,15(0,04)	23,62(0,04)	22,91(0,01)	21,85(0,02)	20,28(0,02)	17,75(0,03)	12,94(0,00)
BankFraud	-0,02 (0,01)	-0,01(0,00)	0,00(0,00)	0,00(0,00)	0,00(0,00)	0,01(0,01)	0,00(0,00)	0,00(0,00)	-0,05(0,02)	-0,02(0,02)	-4,47(0,00)
Census	24,34 (0,13)	24,61(0,17)	25,23(0,27)	25,76(0,08)	26,40(0,09)	26,98(0,09)	27,84(0,31)	28,93(0,12)	30,21(0,08)	31,00(0,22)	24,66(0,00)
Coil2000	-2,17 (0,28)	-2,21(0,14)	-1,93(0,08)	-1,64(0,08)	-1,96(0,35)	-1,87(0,37)	-2,10(0,29)	-2,16(0,70)	-3,13(0,16)	-5,11(0,14)	-91,29(0,00)
Connect-4	35,66 (0,14)	35,92(0,11)	36,15(0,15)	36,43(0,21)	36,56(0,10)	36,90(0,17)	37,43(0,13)	38,18(0,02)	39,74(0,07)	42,55(0,17)	41,01(0,00)
CovType	89,68 (0,02)	89,71(0,03)	89,75(0,02)	89,84(0,04)	89,94(0,03)	90,04(0,05)	90,19(0,05)	90,33(0,03)	90,59(0,05)	91,10(0,03)	91,90(0,00)
Electricity	77,84 (0,13)	77,95(0,04)	77,95(0,20)	78,11(0,16)	78,21(0,11)	78,23(0,11)	78,24(0,03)	78,45(0,04)	78,72(0,07)	78,81(0,20)	80,19(0,00)
Gassensor	87,55 (0,12)	86,64(0,14)	86,81(0,09)	87,03(0,10)	86,47(0,17)	86,14(0,20)	84,78(0,42)	83,50(0,13)	84,47(0,24)	85,72(0,22)	87,72(0,00)
GMSC	3,80 (0,22)	3,77(0,05)	3,84(0,27)	3,80(0,06)	3,63(0,29)	3,64(0,19)	3,41(0,29)	3,13(0,29)	2,33(0,14)	0,46(0,18)	-9,07(0,00)
Kddcup99	99,67 (0,01)	99,76(0,00)	99,73(0,01)	99,72(0,00)	99,74(0,01)	99,73(0,01)	99,69(0,01)	99,69(0,00)	99,64(0,01)	99,57(0,02)	98,92(0,00)
Keystroke	82,67 (0,34)	83,60(0,55)	83,24(0,40)	83,51(0,58)	83,67(0,56)	84,31(1,05)	83,64(0,29)	83,91(0,33)	83,69(0,88)	83,24(0,76)	83,37(0,00)
NOAA	34,02 (0,09)	34,06(0,19)	34,25(0,16)	33,82(0,36)	33,55(0,51)	33,50(0,41)	33,44(0,07)	33,05(0,71)	32,77(0,54)	31,35(0,24)	27,76(0,00)
Nomao	90,36 (0,18)	90,26(0,06)	90,19(0,14)	90,38(0,05)	90,27(0,07)	90,05(0,12)	90,12(0,04)	90,23(0,06)	90,10(0,17)	90,48(0,12)	91,77(0,00)
Outdoor	62,77 (0,55)	64,27(0,28)	62,83(0,27)	62,48(0,11)	62,48(0,17)	63,06(0,03)	63,89(0,11)	64,40(0,05)	65,16(0,16)	66,11(0,15)	69,38(0,00)
Ozone	-0,15 (0,81)	0,63(0,88)	0,62(0,51)	1,46(0,78)	3,59(0,81)	3,12(0,51)	5,62(0,51)	5,21(1,28)	5,83(1,56)	-0,41(1,79)	-26,87(0,00)
Poker	77,51 (0,06)	77,83(0,06)	78,20(0,06)	78,55(0,06)	79,04(0,05)	79,58(0,04)	80,45(0,03)	81,45(0,09)	82,68(0,12)	84,27(0,13)	84,69(0,00)
Rialto	69,13 (0,16)	69,01(0,07)	68,86(0,17)	68,49(0,04)	68,19(0,17)	67,84(0,15)	67,22(0,13)	67,14(0,20)	67,63(0,33)	68,33(0,12)	69,49(0,00)
AGR	81,92 (0,02)	82,19(0,05)	82,41(0,01)	82,58(0,02)	82,74(41,37)	82,84(0,02)	82,89(0,06)	82,92(0,05)	82,80(0,02)	82,26(0,04)	79,98(0,00)
AGR_a	74,24 (0,17)	74,22(0,11)	73,93(0,13)	73,50(0,27)	73,21(0,17)	72,65(0,07)	71,20(0,20)	70,30(0,11)	69,34(0,50)	67,15(0,10)	68,72(0,00)
AGR_g	62,69 (0,13)	63,02(0,21)	63,20(0,28)	62,94(0,13)	62,61(0,12)	62,24(0,05)	61,77(0,31)	61,03(0,26)	60,42(0,19)	60,04(0,30)	62,35(0,00)
HYPER	71,42 (0,04)	71,54(0,06)	71,62(0,04)	71,74(0,02)	71,97(0,01)	72,14(0,03)	72,40(0,06)	72,74(0,01)	73,16(0,07)	73,05(0,04)	70,43(0,00)
LED	70,98 (0,01)	70,98(0,01)	70,97(0,02)	70,96(0,00)	70,99(35,50)	70,97(0,01)	70,98(0,02)	70,98(0,02)	70,96(0,01)	70,96(0,01)	69,36(0,00)
LED_a	71,01 (0,00)	71,00(0,01)	71,02(0,02)	71,02(0,01)	70,99(0,01)	71,00(0,01)	70,97(0,01)	70,95(0,01)	70,91(0,01)	70,79(0,02)	70,37(0,00)
LED_g	70,08 (0,02)	70,07(0,01)	70,06(0,01)	70,07(0,00)	70,05(0,01)	70,04(0,01)	70,04(0,02)	70,00(0,01)	69,98(0,01)	69,95(0,02)	69,35(0,00)
RBF_f	66,11 (0,04)	65,89(0,02)	65,62(0,00)	65,23(0,03)	64,83(0,01)	64,35(0,04)	64,12(0,05)	64,22(0,02)	64,55(0,01)	64,84(0,04)	64,94(0,00)
RBF_m	81,67 (0,02)	81,73(0,02)	81,80(0,02)	81,82(0,02)	81,86(0,03)	81,93(0,02)	81,94(0,01)	81,95(0,01)	81,96(0,02)	81,93(0,03)	81,99(0,00)
RTG	90,73 (0,13)	90,22(0,18)	89,79(0,12)	89,51(0,26)	89,06(0,15)	88,63(0,14)	88,10(0,27)	88,11(0,15)	88,21(0,08)	88,17(0,08)	86,97(0,00)
SEA	71,06 (0,01)	71,08(0,01)	71,07(0,02)	71,03(0,02)	71,00(0,01)	70,94(0,02)	70,83(0,00)	70,70(0,01)	70,48(0,02)	70,01(0,02)	67,76(0,00)
SEA_a	75,58 (0,02)	75,57(0,01)	75,53(0,02)	75,45(0,01)	75,36(0,04)	75,27(0,01)	75,18(0,02)	75,05(0,01)	74,80(0,02)	74,25(0,02)	72,22(0,00)
SEA_g	73,46 (1,65)	74,43(0,02)	74,43(0,01)	74,44(0,02)	74,40(0,02)	74,35(0,02)	74,26(0,01)	74,09(0,01)	73,86(0,01)	73,33(0,02)	71,44(0,00)
SINE	98,97 (0,01)	98,96(0,00)	98,96(0,02)	98,98(0,01)	98,96(0,01)	98,90(0,01)	98,89(0,01)	98,84(0,01)	98,80(0,01)	98,68(0,01)	98,30(0,00)
WAVEFORM	78,14 (0,04)	78,24(0,01)	78,28(0,01)	78,34(0,02)	78,41(0,02)	78,44(0,01)	78,48(0,00)	78,50(0,04)	78,52(0,01)	78,41(0,03)	78,16(0,00)
WAVEFORM_g	77,74 (0,00)	77,80(0,02)	77,85(0,04)	77,86(0,02)	77,93(0,02)	77,96(0,03)	78,03(0,04)	78,03(0,02)	78,02(0,01)	77,95(0,00)	77,73(0,00)
Ranking médio geral	6,12	5,42	5,24	4,94	5,85	5,45	5,79	5,73	6,39	7,09	7,70

Table 14: Comparison of the accuracy metric between the ARTE algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARTE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	67.45 (0.18)	63.68(0.00)	63.59(0.04)	63.54(0.00)	67.12(0.03)	63.15(0.03)	62.79(0.01)	62.36(0.00)	61.80(0.00)	60.87(0.02)	59.01(0.00)
BankFraud	98.89 (0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.21(0.00)
Census	95.69 (0.05)	95.38(0.00)	95.39(0.00)	95.42(0.01)	95.84(0.01)	95.49(0.01)	95.54(0.01)	95.59(0.01)	95.69(0.00)	95.74(0.01)	95.03(0.00)
Coil2000	93.94 (0.05)	93.93(0.00)	93.91(0.00)	93.92(0.01)	93.92(0.03)	93.93(0.00)	93.93(0.01)	93.92(0.04)	93.83(0.02)	93.51(0.05)	88.53(0.00)
Connect-4	78.30 (0.36)	78.19(0.01)	78.17(0.07)	78.28(0.07)	78.41(0.04)	78.59(0.03)	78.75(0.06)	79.01(0.01)	79.29(0.01)	80.07(0.05)	80.20(0.00)
CovType	94.96 (0.07)	95.28(0.01)	95.32(0.00)	95.36(0.01)	95.37(0.01)	95.39(0.01)	95.45(0.01)	95.50(0.01)	95.60(0.00)	95.79(0.00)	96.05(0.00)
Electricity	91.13 (0.23)	90.84(0.04)	90.83(0.03)	90.82(0.04)	90.93(0.06)	91.03(0.01)	91.04(0.03)	91.11(0.06)	91.32(0.03)	91.62(0.01)	91.91(0.00)
Gassensor	95.58 (0.22)	95.60(0.08)	95.69(0.12)	95.65(0.10)	95.70(0.05)	95.42(0.01)	95.41(0.07)	95.50(0.02)	95.40(0.12)	95.16(0.23)	94.85(0.00)
GMSC	93.38 (0.02)	93.57(0.00)	93.56(0.01)	93.55(0.00)	93.57(0.01)	93.55(0.01)	93.54(0.01)	93.50(0.02)	93.42(0.01)	93.24(0.00)	91.70(0.00)
Kddcup99	99.78 (0.04)	99.89(0.00)	99.90(0.01)	99.90(0.00)	99.90(0.00)	99.90(0.01)	99.90(0.01)	99.90(0.00)	99.90(0.00)	99.90(0.01)	99.88(0.00)
Keystroke	94.04 (0.29)	94.19(0.06)	94.28(0.16)	94.03(0.16)	94.02(0.23)	94.19(0.13)	93.81(0.00)	94.00(0.13)	93.56(0.06)	92.75(0.31)	93.06(0.00)
NOAA	79.08 (0.00)	79.37(0.11)	79.41(0.13)	79.41(0.07)	79.33(0.08)	79.28(0.11)	79.40(0.05)	79.20(0.12)	78.78(0.04)	78.37(0.07)	76.89(0.00)
Nomao	97.40 (0.04)	97.41(0.03)	97.31(0.02)	97.34(0.01)	97.36(0.04)	97.34(0.04)	97.34(0.00)	97.35(0.02)	97.40(0.02)	97.52(0.03)	97.52(0.00)
Outdoor	71.97 (1.23)	74.06(0.01)	73.60(0.30)	72.83(0.27)	72.98(0.46)	72.56(0.04)	72.71(0.06)	73.25(0.30)	74.00(0.15)	74.64(0.66)	75.47(0.00)
Ozone	93.83 (0.13)	93.84(0.04)	93.92(0.08)	93.90(0.02)	93.86(0.14)	93.98(0.10)	93.92(0.00)	93.98(0.06)	93.94(0.02)	93.66(0.02)	91.91(0.00)
Poker	94.43 (1.42)	87.41(0.05)	87.13(0.06)	86.83(0.01)	94.17(0.03)	86.21(0.02)	85.91(0.06)	85.43(0.04)	85.63(0.08)	86.37(0.12)	86.61(0.00)
Rialto	80.78 (0.40)	80.13(0.00)	79.97(0.01)	79.85(0.08)	80.33(0.38)	79.37(0.01)	79.19(0.05)	79.02(0.04)	78.70(0.02)	78.83(0.02)	78.73(0.00)
AGR	94.99 (0.04)	92.93(0.09)	93.13(0.01)	93.42(0.07)	94.91(0.00)	93.70(0.01)	93.82(0.03)	93.86(0.02)	93.94(0.01)	93.79(0.01)	91.85(0.00)
AGR_a	94.40 (0.02)	79.74(0.14)	80.05(0.04)	80.41(0.00)	94.33(0.01)	80.79(0.03)	81.16(0.05)	80.82(0.03)	80.35(0.08)	80.01(0.02)	79.04(0.00)
AGR_g	91.86 (0.02)	75.64(0.08)	76.14(0.07)	76.51(0.12)	91.78(0.01)	76.79(0.05)	76.91(0.03)	76.72(0.09)	76.41(0.17)	76.30(0.05)	75.38(0.00)
HYPER	87.52 (0.19)	86.99(0.01)	87.10(0.00)	87.22(0.01)	87.37(0.03)	87.54(0.00)	87.75(0.01)	88.03(0.03)	88.34(0.01)	88.50(0.04)	86.01(0.00)
LED	73.50 (0.71)	73.92(0.02)	73.92(0.00)	73.89(0.01)	73.92(0.00)	73.93(0.02)	73.93(0.02)	73.93(0.01)	73.91(0.00)	73.87(0.00)	72.53(0.00)
LED_a	73.04 (0.93)	73.91(0.00)	73.92(0.01)	73.91(0.00)	73.93(0.02)	73.93(0.02)	73.92(0.00)	73.93(0.01)	73.91(0.02)	73.83(0.01)	72.92(0.00)
LED_g	71.93 (1.03)	72.98(0.00)	72.94(0.01)	72.93(0.01)	72.94(0.02)	72.95(0.01)	72.95(0.00)	72.94(0.02)	72.97(0.00)	72.98(0.01)	71.89(0.00)
RBF_f	79.27 (0.34)	79.65(0.01)	79.63(0.00)	79.56(0.04)	79.33(0.02)	78.94(0.02)	78.35(0.00)	77.83(0.02)	77.71(0.03)	77.66(0.02)	76.43(0.00)
RBF_m	87.55 (0.08)	87.71(0.00)	87.83(0.02)	87.85(0.00)	87.93(0.01)	88.03(0.01)	88.13(0.00)	88.24(0.00)	88.26(0.01)	88.24(0.00)	87.96(0.00)
RTG	96.89 (0.29)	93.17(0.07)	93.11(0.06)	93.28(0.02)	96.73(0.06)	93.30(0.06)	93.44(0.00)	93.63(0.11)	93.73(0.13)	94.15(0.16)	92.97(0.00)
SEA	89.87 (0.00)	89.88(0.00)	89.87(0.00)	89.87(0.00)	89.86(0.00)	89.86(0.00)	89.84(0.00)	89.83(0.01)	89.78(0.00)	89.57(0.00)	84.52(0.00)
SEA_a	89.73 (0.01)	89.74(0.00)	89.74(0.00)	89.73(0.00)	89.72(0.00)	89.70(0.00)	89.68(0.01)	89.65(0.01)	89.57(0.00)	89.28(0.00)	84.51(0.00)
SEA_g	89.31 (0.01)	89.32(0.00)	89.31(0.00)	89.31(0.01)	89.30(0.00)	89.29(0.00)	89.27(0.00)	89.24(0.00)	89.17(0.01)	88.90(0.01)	84.09(0.00)
SINE	99.88 (0.00)	99.88(0.00)	99.88(0.00)	99.88(0.00)	99.88(0.00)	99.88(0.00)	99.87(0.00)	99.87(0.00)	99.87(0.01)	99.86(0.01)	99.81(0.00)
WAVEFORM	85.70 (0.05)	85.93(0.01)	85.95(0.01)	85.95(0.01)	85.97(0.01)	85.99(0.01)	85.99(0.00)	85.95(0.02)	85.87(0.01)	85.73(0.01)	85.39(0.00)
WAVEFORM_g	85.39 (0.06)	85.68(0.02)	85.71(0.01)	85.73(0.02)	85.74(0.01)	85.74(0.01)	85.73(0.03)	85.69(0.02)	85.66(0.00)	85.48(0.01)	85.16(0.00)
Ranking médio geral	5,36	5,18	5,33	5,45	5,12	4,79	5,21	5,24	6,06	6,61	9,00

Table 15: Comparison of the Kappa M metric between the ARTE algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	ARTE	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	26.94 (0.40)	18.46(0.00)	18.26(0.09)	18.16(0.00)	26.20(0.07)	17.28(0.06)	16.48(0.03)	15.50(0.00)	14.25(0.01)	12.16(0.04)	7.98(0.00)
BankFraud	0.00 (0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.01(0.01)	0.00(0.00)	-0.03(0.04)	-0.24(0.05)	-61.98(0.00)
Census	30.64 (0.87)	25.55(0.11)	25.77(0.05)	26.23(0.06)	33.03(0.11)	27.43(0.19)	28.19(0.19)	28.90(0.09)	30.46(0.08)	31.42(0.16)	19.89(0.00)
Coil2000	-1.50 (0.89)	-1.70(0.00)	-2.04(0.00)	-1.96(0.09)	-1.91(0.53)	-1.70(0.00)	-1.79(0.09)	-1.96(0.60)	-3.33(0.26)	-8.70(0.86)	-92.15(0.00)
Connect-4	36.51 (1.05)	36.19(0.04)	36.11(0.18)	36.45(0.22)	36.81(0.10)	37.35(0.09)	37.83(0.17)	38.56(0.01)	39.40(0.03)	41.69(0.15)	42.07(0.00)
CovType	90.16 (0.13)	90.79(0.02)	90.85(0.01)	90.94(0.00)	90.96(0.02)	91.02(0.02)	91.11(0.03)	91.22(0.01)	91.42(0.00)	91.78(0.00)	92.30(0.00)
Electricity	79.10 (0.53)	78.42(0.12)	78.40(0.08)	78.38(0.09)	78.65(0.13)	78.87(0.02)	78.90(0.08)	79.06(0.14)	79.56(0.07)	80.26(0.02)	80.94(0.00)
Gassensor	94.24 (0.28)	94.26(0.09)	94.36(0.15)	94.31(0.14)	94.38(0.06)	94.02(0.02)	94.01(0.09)	94.12(0.02)	94.00(0.16)	93.68(0.31)	93.27(0.00)
GMSC	1.00 (0.20)	3.87(0.01)	3.66(0.10)	3.49(0.13)	3.84(0.21)	3.62(0.16)	3.41(0.08)	2.81(0.25)	1.62(0.10)	-1.05(0.05)	-24.04(0.00)
Kddcup99	99.48 (0.09)	99.76(0.00)	99.77(0.00)	99.77(0.00)	99.76(0.00)	99.76(0.00)	99.77(0.00)	99.77(0.00)	99.77(0.00)	99.76(0.01)	99.72(0.00)
Keystroke	87.21 (0.61)	87.53(0.13)	87.73(0.34)	87.20(0.34)	87.18(0.49)	87.53(0.27)	86.72(0.00)	87.13(0.27)	86.19(0.13)	84.45(0.67)	85.12(0.00)
NOAA	33.34 (0.01)	34.26(0.36)	34.37(0.40)	34.39(0.21)	34.13(0.25)	33.99(0.35)	34.36(0.18)	33.70(0.36)	32.38(0.15)	31.08(0.21)	26.36(0.00)
Nomao	90.90 (0.14)	90.94(0.09)	90.61(0.07)	90.69(0.00)	90.77(0.13)	90.68(0.15)	90.69(0.00)	90.73(0.07)	90.90(0.05)	91.31(0.10)	91.32(0.00)
Outdoor	71.18 (1.27)	73.33(0.02)	72.85(0.31)	72.05(0.28)	72.21(0.47)	71.78(0.04)	71.94(0.06)	72.49(0.31)	73.26(0.15)	73.92(0.68)	74.78(0.00)
Ozone	2.37 (2.06)	2.50(0.63)	3.75(1.25)	3.44(0.32)	2.75(2.25)	4.69(1.57)	3.75(0.00)	4.69(0.93)	4.06(0.31)	-0.31(0.31)	-28.12(0.00)
Poker	88.84 (2.85)	74.76(0.10)	74.20(0.13)	73.61(0.02)	88.33(0.07)	72.35(0.03)	71.75(0.13)	70.81(0.09)	71.20(0.16)	72.68(0.24)	73.16(0.00)
Rialto	78.64 (0.44)	77.92(0.00)	77.74(0.00)	77.61(0.09)	78.15(0.43)	77.08(0.01)	76.88(0.06)	76.69(0.04)	76.33(0.02)	76.48(0.02)	76.37(0.00)
AGR	84.73 (0.12)	78.44(0.28)	79.06(0.02)	79.93(0.19)	84.50(0.00)	80.80(0.04)	81.15(0.10)	81.27(0.05)	81.53(0.04)	81.08(0.02)	75.15(0.00)
AGR_a	88.13 (0.04)	57.05(0.29)	57.71(0.09)	58.47(0.01)	87.99(0.02)	59.26(0.07)	60.07(0.09)	59.34(0.06)	58.35(0.18)	57.62(0.04)	55.57(0.00)
AGR_g	82.75 (0.05)	48.37(0.17)	49.43(0.15)	50.20(0.25)	82.56(0.02)	50.80(0.11)	51.06(0.05)	50.65(0.19)	50.00(0.36)	49.76(0.09)	47.82(0.00)
HYPER	75.00 (0.39)	73.95(0.02)	74.17(0.00)	74.40(0.02)	74.71(0.06)	75.04(0.00)	75.47(0.02)	76.03(0.05)	76.64(0.01)	76.96(0.08)	71.97(0.00)
LED	70.53 (0.79)	70.99(0.02)	71.00(0.01)	70.96(0.01)	71.00(0.00)	71.01(0.02)	71.00(0.02)	71.00(0.01)	70.99(0.00)	70.94(0.00)	69.45(0.00)
LED_a	70.00 (1.04)	70.97(0.01)	70.98(0.01)	70.97(0.00)	70.98(0.02)	70.99(0.02)	70.98(0.01)	70.98(0.01)	70.96(0.02)	70.88(0.02)	69.87(0.00)
LED_g	68.76 (1.15)	69.93(0.01)	69.89(0.02)	69.87(0.01)	69.88(0.02)	69.90(0.02)	69.89(0.00)	69.88(0.02)	69.92(0.00)	69.93(0.01)	68.71(0.00)
RBF_f	70.38 (0.49)	70.92(0.01)	70.89(0.01)	70.79(0.05)	70.47(0.03)	69.91(0.03)	69.06(0.01)	68.32(0.02)	68.15(0.04)	68.08(0.02)	66.33(0.00)
RBF_m	82.21 (0.11)	82.44(0.00)	82.61(0.02)	82.65(0.00)	82.76(0.02)	82.90(0.02)	83.05(0.00)	83.20(0.01)	83.23(0.01)	83.20(0.00)	82.80(0.00)
RTG	92.63 (0.70)	83.79(0.16)	83.66(0.14)	84.06(0.06)	92.25(0.15)	84.12(0.13)	84.45(0.00)	84.89(0.25)	85.12(0.32)	86.13(0.37)	83.32(0.00)
SEA	71.60 (0.01)	71.62(0.01)	71.63(0.00)	71.61(0.00)	71.59(0.00)	71.58(0.01)	71.53(0.00)	71.50(0.01)	71.37(0.00)	70.76(0.01)	56.62(0.00)
SEA_a	76.20 (0.01)	76.23(0.01)	76.21(0.01)	76.19(0.00)	76.18(0.01)	76.14(0.01)	76.08(0.01)	76.00(0.01)	75.82(0.00)	75.17(0.01)	64.11(0.00)
SEA_g	75.23 (0.02)	75.24(0.01)	75.24(0.00)	75.22(0.01)	75.20(0.01)	75.18(0.01)	75.13(0.01)	75.07(0.00)	74.89(0.00)	74.28(0.02)	63.13(0.00)
SINE	99.74 (0.00)	99.74(0.00)	99.74(0.00)	99.75(0.01)	99.74(0.00)	99.74(0.00)	99.73(0.00)	99.73(0.00)	99.72(0.01)	99.70(0.01)	99.60(0.00)
WAVEFORM	78.51 (0.08)	78.85(0.01)	78.88(0.01)	78.88(0.00)	78.92(0.02)	78.95(0.01)	78.94(0.01)	78.89(0.04)	78.76(0.01)	78.55(0.01)	78.04(0.00)
WAVEFORM_g	78.05 (0.09)	78.49(0.03)	78.53(0.02)	78.56(0.03)	78.58(0.02)	78.58(0.02)	78.56(0.04)	78.50(0.02)	78.45(0.01)	78.18(0.01)	77.70(0.00)
Ranking médio geral	5,42	5,36	5,33	5,55	5,42	4,91	5,15	5,33	6,27	6,97	9,00

Table 16: Comparison of the accuracy metric between the LB algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	LB	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	63,48 (0,03)	63.51(0.02)	63.48(0.01)	63.42(0.02)	63.30(0.01)	63.09(0.04)	62.75(0.03)	62.26(0.01)	61.56(0.02)	60.46(0.01)	58.92(0.00)
BankFraud	98,89 (0,00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.88(0.00)	97.61(0.00)
Census	95,17 (0,01)	95.32(0.00)	95.36(0.01)	95.38(0.01)	95.43(0.00)	95.50(0.02)	95.60(0.02)	95.75(0.01)	95.93(0.00)	96.05(0.01)	95.52(0.00)
Coil2000	93,90 (0,01)	93.85(0.00)	93.88(0.01)	93.94(0.00)	93.92(0.02)	93.94(0.02)	93.91(0.01)	93.87(0.01)	93.87(0.01)	93.72(0.02)	82.78(0.00)
Connect-4	79,04 (0,07)	77.88(0.00)	77.87(0.03)	78.07(0.05)	78.20(0.02)	78.25(0.04)	78.46(0.07)	78.75(0.05)	79.28(0.03)	79.70(0.07)	78.90(0.00)
CovType	93,38 (0,03)	91.90(0.02)	91.76(0.01)	91.77(0.02)	91.70(0.01)	91.78(0.05)	91.89(0.02)	92.20(0.02)	92.70(0.02)	93.29(0.04)	94.53(0.00)
Electricity	89,16 (0,02)	90.27(0.07)	90.22(0.03)	90.40(0.01)	90.38(0.05)	90.46(0.01)	90.52(0.05)	90.78(0.02)	91.18(0.08)	91.97(0.07)	93.27(0.00)
Gassensor	80,71 (0,34)	83.51(0.33)	83.44(0.02)	84.24(0.22)	84.49(0.13)	84.49(0.11)	84.69(0.21)	84.56(0.09)	84.43(0.18)	84.14(0.12)	84.98(0.00)
GMSC	93,53 (0,01)	93.54(0.00)	93.54(0.02)	93.56(0.01)	93.53(0.02)	93.53(0.01)	93.47(0.01)	93.39(0.03)	93.36(0.02)	93.13(0.01)	91.97(0.00)
Kddcup99	99,80 (0,00)	99.80(0.00)	99.80(0.00)	99.80(0.00)	99.81(0.00)	99.81(0.00)	99.81(0.00)	99.80(0.00)	99.80(0.00)	99.79(0.00)	99.82(0.00)
Keystroke	87,53 (0,13)	87.56(0.36)	87.81(0.27)	87.81(0.05)	87.39(0.63)	87.20(0.21)	86.91(0.11)	86.00(0.36)	84.93(0.10)	83.95(0.23)	82.43(0.00)
NOAA	78,52 (0,06)	78.14(0.04)	78.20(0.13)	78.09(0.10)	78.23(0.05)	78.14(0.03)	78.09(0.17)	78.15(0.07)	78.26(0.10)	77.14(0.09)	77.13(0.00)
Nomao	96,25 (0,06)	96.41(0.03)	96.37(0.03)	96.50(0.03)	96.53(0.03)	96.55(0.02)	96.53(0.03)	96.60(0.02)	96.51(0.03)	96.57(0.05)	97.04(0.00)
Outdoor	57,52 (0,14)	60.30(0.04)	60.47(0.13)	61.35(0.30)	61.35(0.13)	61.81(0.17)	61.60(0.37)	62.36(0.23)	62.94(0.33)	63.83(0.19)	67.57(0.00)
Ozone	93,57 (0,10)	93.35(0.07)	93.47(0.04)	93.43(0.08)	93.71(0.05)	93.53(0.04)	93.61(0.02)	93.71(0.02)	93.69(0.02)	92.85(0.09)	89.14(0.00)
Poker	95,47 (0,08)	86.12(0.22)	84.91(0.24)	83.26(0.08)	82.12(0.16)	81.14(0.19)	80.80(0.03)	80.80(0.07)	81.48(0.09)	83.17(0.08)	84.20(0.00)
Rialto	61,09 (0,10)	60.86(0.07)	60.72(0.09)	60.30(0.03)	59.96(0.16)	59.92(0.08)	59.72(0.09)	59.45(0.12)	59.18(0.10)	58.36(0.09)	57.46(0.00)
AGR	94,77 (0,01)	94.78(0.01)	94.75(0.01)	94.72(0.00)	94.68(0.01)	94.64(0.00)	94.59(0.01)	94.53(0.01)	94.46(0.01)	94.28(0.01)	92.96(0.00)
AGR_a	89,70 (0,21)	90.61(0.05)	90.41(0.07)	89.96(0.11)	89.51(0.05)	88.25(0.53)	86.29(0.81)	84.71(0.46)	83.95(0.14)	83.60(0.09)	86.75(0.00)
AGR_g	86,10 (0,03)	85.21(0.54)	84.14(0.23)	82.77(0.21)	82.03(0.08)	81.71(0.22)	81.27(0.10)	80.96(0.10)	80.79(0.02)	80.35(0.07)	80.24(0.00)
HYPER	87,56 (0,01)	87.99(0.01)	88.19(0.02)	88.45(0.01)	88.75(0.01)	89.08(0.00)	89.45(0.00)	89.82(0.01)	90.18(0.02)	90.52(0.02)	88.60(0.00)
LED	73,94 (0,00)	73.95(0.00)	73.96(0.01)	73.97(0.01)	73.96(0.00)	73.95(0.00)	73.93(0.01)	73.57(0.00)	72.80(0.00)	71.97(0.00)	71.98(0.00)
LED_a	73,91 (0,04)	74.07(0.00)	74.07(0.01)	74.08(0.00)	74.08(0.01)	74.07(0.01)	74.02(0.00)	73.70(0.00)	72.93(0.01)	72.12(0.02)	72.09(0.00)
LED_g	73,23 (0,01)	73.17(0.00)	73.17(0.01)	73.17(0.00)	73.16(0.00)	73.12(0.00)	73.01(0.00)	72.57(0.00)	71.78(0.01)	70.97(0.02)	70.99(0.00)
RBF_f	59,83 (0,00)	55.24(0.02)	55.00(0.01)	54.91(0.00)	54.92(0.03)	55.03(0.00)	55.13(0.01)	54.65(0.02)	53.61(0.00)	52.90(0.03)	52.27(0.00)
RBF_m	85,09 (0,02)	85.36(0.03)	85.21(0.01)	84.93(0.06)	84.40(0.03)	83.25(0.07)	80.81(0.09)	75.76(0.11)	70.79(0.20)	73.88(0.08)	77.15(0.00)
RTG	97,84 (0,01)	97.85(0.00)	97.82(0.01)	97.75(0.03)	97.66(0.01)	97.54(0.01)	97.42(0.01)	97.25(0.01)	96.99(0.03)	96.69(0.02)	96.17(0.00)
SEA	89,86 (0,00)	89.87(0.00)	89.87(0.00)	89.87(0.00)	89.86(0.00)	89.85(0.00)	89.84(0.00)	89.82(0.00)	89.78(0.00)	89.20(0.01)	81.74(0.00)
SEA_a	88,61 (0,02)	89.47(0.02)	89.50(0.01)	89.52(0.01)	89.54(0.01)	89.59(0.01)	89.62(0.00)	89.62(0.00)	89.51(0.00)	89.00(0.01)	82.12(0.00)
SEA_g	88,90 (0,01)	89.11(0.00)	89.14(0.00)	89.15(0.01)	89.16(0.00)	89.17(0.00)	89.19(0.00)	89.20(0.00)	89.16(0.00)	88.64(0.01)	81.67(0.00)
SINE	99,84 (0,01)	99.86(0.00)	99.88(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.88(0.00)	99.87(0.00)	99.84(0.00)	99.72(0.00)
WAVEFORM	85,99 (0,01)	86.05(0.00)	86.09(0.00)	86.09(0.01)	86.10(0.00)	86.11(0.01)	86.08(0.01)	86.03(0.01)	85.97(0.01)	85.79(0.02)	82.68(0.00)
WAVEFORM_g	85,86 (0,01)	85.91(0.01)	85.93(0.00)	85.96(0.00)	85.98(0.01)	85.94(0.01)	85.97(0.01)	85.94(0.01)	85.87(0.00)	85.72(0.02)	83.45(0.00)
Ranking médio geral	5,48	5,21	5,15	4,67	4,67	4,85	5,27	5,82	6,76	8,18	8,18

Table 17: Comparison of the Kappa M metric between the LB algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	LB		δ								Inc.
			0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
Airlines	18,05 (0,05)	18.09(0.04)	18.02(0.02)	17.89(0.05)	17.61(0.03)	17.14(0.08)	16.39(0.05)	15.28(0.03)	13.70(0.04)	11.23(0.02)	7.79(0.00)
BankFraud	-0,19 (0,00)	-0.08(0.01)	-0.02(0.01)	-0.09(0.00)	-0.14(0.01)	-0.02(0.01)	-0.06(0.02)	-0.12(0.01)	-0.08(0.02)	-0.88(0.06)	-115.99(0.00)
Census	22,21 (0,09)	24.75(0.08)	25.23(0.12)	25.59(0.19)	26.34(0.15)	27.62(0.27)	29.13(0.25)	31.54(0.18)	34.50(0.07)	36.52(0.14)	27.87(0.00)
Coil2000	-2,30 (0,15)	-3.13(0.08)	-2.56(0.25)	-1.59(0.08)	-1.94(0.28)	-1.59(0.29)	-2.10(0.21)	-2.67(0.22)	-2.67(0.22)	-5.06(0.43)	-188.56(0.00)
Connect-4	38,69 (0,21)	35.26(0.02)	35.25(0.08)	35.83(0.13)	36.12(0.11)	36.36(0.11)	36.97(0.19)	37.83(0.14)	39.36(0.11)	40.60(0.21)	38.27(0.00)
CovType	87,08 (0,05)	84.20(0.05)	83.93(0.01)	83.94(0.04)	83.83(0.04)	83.96(0.10)	84.17(0.04)	84.77(0.04)	85.76(0.04)	86.91(0.07)	89.33(0.00)
Electricity	74,47 (0,05)	77.08(0.17)	76.95(0.06)	77.38(0.02)	77.25(0.14)	77.52(0.03)	77.66(0.13)	78.29(0.04)	79.22(0.19)	81.10(0.17)	84.15(0.00)
Gassensor	74,82 (0,45)	78.47(0.43)	78.37(0.02)	79.42(0.29)	79.82(0.18)	79.75(0.14)	80.01(0.28)	79.84(0.12)	79.66(0.23)	79.30(0.16)	80.38(0.00)
GMSC	3,14 (0,07)	3.37(0.12)	3.43(0.23)	3.79(0.15)	3.29(0.26)	3.29(0.15)	2.40(0.18)	1.20(0.41)	0.67(0.28)	-2.67(0.10)	-20.00(0.00)
Kddcup99	99,55 (0,00)	99.55(0.00)	99.54(0.00)	99.55(0.00)	99.55(0.00)	99.56(0.00)	99.56(0.00)	99.55(0.00)	99.54(0.00)	99.52(0.00)	99.60(0.00)
Keystroke	73,25 (0,28)	73.32(0.76)	73.86(0.57)	73.86(0.11)	73.00(1.25)	72.56(0.44)	71.93(0.23)	69.97(0.76)	67.69(0.22)	65.59(0.49)	62.33(0.00)
NOAA	31,57 (0,19)	30.34(0.13)	30.54(0.40)	30.19(0.33)	30.61(0.16)	30.33(0.08)	30.19(0.55)	30.40(0.23)	30.73(0.32)	27.16(0.30)	27.13(0.00)
Nomao	87,00 (0,23)	87.45(0.12)	87.32(0.09)	87.76(0.11)	87.78(0.11)	87.94(0.07)	87.85(0.11)	88.11(0.07)	87.79(0.09)	88.01(0.18)	89.63(0.00)
Outdoor	56,31 (0,14)	59.17(0.04)	59.35(0.13)	60.25(0.31)	60.32(0.14)	60.72(0.18)	60.52(0.38)	61.29(0.24)	61.89(0.33)	62.81(0.19)	66.65(0.00)
Ozone	-1,71 (1,56)	-5.20(1.18)	-3.32(0.59)	-3.95(1.29)	0.37(0.64)	-2.29(0.59)	-1.03(0.29)	0.41(0.29)	0.21(0.29)	-13.12(1.35)	-71.87(0.00)
Poker	90,93 (0,16)	72.17(0.44)	69.77(0.47)	66.45(0.16)	64.04(0.33)	62.19(0.38)	61.52(0.05)	61.52(0.14)	62.88(0.17)	66.27(0.15)	68.34(0.00)
Rialto	56,76 (0,10)	56.51(0.07)	56.35(0.10)	55.89(0.03)	55.56(0.15)	55.47(0.09)	55.25(0.09)	54.94(0.13)	54.65(0.11)	53.73(0.10)	52.73(0.00)
AGR	84,07 (0,04)	84.08(0.03)	84.00(0.02)	83.90(0.02)	83.81(0.02)	83.67(0.01)	83.52(0.04)	83.34(0.02)	83.11(0.02)	82.58(0.04)	78.54(0.00)
AGR_a	78,16 (0,44)	80.10(0.10)	79.67(0.15)	78.72(0.24)	77.89(0.30)	75.09(1.11)	70.94(1.71)	67.59(0.98)	65.98(0.29)	65.23(0.18)	71.92(0.00)
AGR_g	70,54 (0,06)	68.65(1.15)	66.39(0.49)	63.49(0.45)	61.99(0.25)	61.24(0.48)	60.29(0.21)	59.65(0.21)	59.28(0.04)	58.34(0.15)	58.12(0.00)
HYPER	75,07 (0,02)	75.93(0.02)	76.35(0.03)	76.87(0.02)	77.48(0.02)	78.14(0.01)	78.86(0.01)	79.60(0.02)	80.34(0.03)	81.00(0.05)	77.16(0.00)
LED	71,02 (0,00)	71.03(0.00)	71.04(0.01)	71.05(0.01)	71.04(0.01)	71.03(0.00)	71.01(0.01)	70.60(0.00)	69.75(0.01)	68.83(0.00)	68.84(0.00)
LED_a	70,97 (0,05)	71.15(0.00)	71.14(0.01)	71.15(0.00)	71.15(0.01)	71.14(0.01)	71.09(0.00)	70.73(0.00)	69.87(0.01)	68.97(0.02)	68.94(0.00)
LED_g	70,20 (0,01)	70.14(0.00)	70.13(0.01)	70.14(0.01)	70.12(0.00)	70.09(0.00)	69.96(0.00)	69.46(0.00)	68.59(0.01)	67.69(0.02)	67.71(0.00)
RBF_f	42,62 (0,01)	36.06(0.03)	35.71(0.02)	35.57(0.01)	35.59(0.04)	35.75(0.01)	35.89(0.02)	35.21(0.03)	33.72(0.01)	32.71(0.05)	31.81(0.00)
RBF_m	78,70 (0,03)	79.08(0.04)	78.88(0.02)	78.47(0.08)	77.74(0.06)	76.07(0.09)	72.59(0.12)	65.38(0.15)	58.26(0.29)	62.69(0.11)	67.36(0.00)
RTG	94,88 (0,03)	94.89(0.01)	94.82(0.03)	94.66(0.07)	94.45(0.03)	94.18(0.03)	93.89(0.02)	93.48(0.02)	92.86(0.07)	92.16(0.04)	90.92(0.00)
SEA	71,58 (0,01)	71.62(0.00)	71.61(0.01)	71.62(0.00)	71.58(0.00)	71.56(0.01)	71.54(0.00)	71.47(0.00)	71.36(0.01)	69.74(0.02)	48.84(0.00)
SEA_a	73,60 (0,03)	75.59(0.04)	75.68(0.04)	75.71(0.04)	75.77(0.03)	75.87(0.01)	75.95(0.01)	75.94(0.00)	75.70(0.01)	74.50(0.03)	58.56(0.00)
SEA_g	74,27 (0,02)	74.77(0.02)	74.83(0.01)	74.86(0.01)	74.86(0.01)	74.89(0.01)	74.96(0.00)	74.97(0.00)	74.87(0.00)	73.68(0.02)	57.52(0.00)
SINE	99,68 (0,01)	99.71(0.00)	99.75(0.00)	99.76(0.00)	99.77(0.00)	99.76(0.00)	99.76(0.00)	99.75(0.00)	99.72(0.00)	99.66(0.01)	99.41(0.00)
WAVEFORM	78,94 (0,01)	79.03(0.00)	79.09(0.01)	79.10(0.01)	79.11(0.01)	79.13(0.03)	79.08(0.01)	79.00(0.02)	78.92(0.02)	78.65(0.03)	73.97(0.00)
WAVEFORM_g	78,75 (0,01)	78.83(0.02)	78.86(0.01)	78.91(0.00)	78.94(0.02)	78.89(0.02)	78.92(0.02)	78.88(0.01)	78.77(0.00)	78.55(0.03)	75.13(0.00)
Ranking médio geral	5,70	5,24	5,39	4,88	4,91	4,88	5,39	6,06	6,88	8,18	8,18

Table 18: Comparison of the accuracy metric between the SRP algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	SRP	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	68.56 (0.00)	68.54(0.02)	68.42(0.01)	68.35(0.02)	68.17(0.01)	67.96(0.03)	67.60(0.01)	67.15(0.05)	66.72(0.01)	65.90(0.01)	63.87(0.00)
BankFraud	98.89 (0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.89(0.00)	98.87(0.00)
Census	95.27 (0.00)	95.30(0.03)	95.34(0.02)	95.36(0.01)	95.40(0.00)	95.45(0.01)	95.53(0.02)	95.59(0.01)	95.66(0.01)	95.71(0.01)	95.43(0.00)
Coil2000	93.95 (0.02)	93.90(0.02)	93.94(0.02)	93.95(0.01)	93.95(0.01)	93.99(0.00)	93.97(0.00)	93.94(0.01)	93.91(0.00)	93.78(0.00)	89.28(0.00)
Connect-4	78.75 (0.06)	78.89(0.02)	78.84(0.01)	79.01(0.04)	79.03(0.02)	79.13(0.06)	79.28(0.02)	79.36(0.04)	79.70(0.01)	80.08(0.01)	80.19(0.00)
CovType	95.33 (0.02)	95.34(0.01)	95.36(0.01)	95.37(0.02)	95.38(0.02)	95.38(0.00)	95.37(0.01)	95.36(0.02)	95.38(0.02)	95.34(0.02)	95.56(0.00)
Electricity	89.81 (0.06)	89.71(0.00)	89.64(0.02)	89.64(0.02)	89.66(0.09)	89.88(0.03)	89.80(0.09)	89.75(0.06)	90.09(0.06)	90.85(0.06)	92.30(0.00)
Gassensor	93.33 (0.10)	93.09(0.13)	93.07(0.02)	92.87(0.12)	92.73(0.07)	92.29(0.40)	92.49(0.21)	93.03(0.07)	93.11(0.15)	92.97(0.15)	92.53(0.00)
GMSC	93.49 (0.01)	93.51(0.01)	93.54(0.01)	93.53(0.01)	93.53(0.02)	93.55(0.00)	93.56(0.01)	93.57(0.01)	93.56(0.01)	93.59(0.01)	93.01(0.00)
Kddcup99	99.88 (0.00)	99.88(0.01)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)	99.89(0.00)
Keystroke	94.57 (0.19)	94.37(0.50)	94.62(0.06)	94.50(0.00)	94.65(0.28)	94.28(0.16)	94.03(0.47)	93.84(0.09)	94.09(0.03)	93.91(0.03)	92.93(0.00)
NOAA	79.23 (0.00)	79.50(0.05)	79.53(0.07)	79.42(0.04)	79.50(0.05)	79.72(0.05)	79.53(0.04)	79.50(0.08)	79.44(0.07)	79.03(0.07)	77.61(0.00)
Nomao	97.38 (0.00)	97.36(0.02)	97.31(0.01)	97.37(0.01)	97.32(0.01)	97.28(0.01)	97.21(0.04)	97.28(0.01)	97.22(0.02)	97.23(0.02)	97.60(0.00)
Outdoor	68.46 (0.31)	68.85(0.05)	68.43(0.63)	68.60(0.13)	68.61(0.24)	68.35(0.03)	68.94(0.13)	68.85(0.30)	68.59(0.18)	69.99(0.18)	70.82(0.00)
Ozone	93.64 (0.07)	93.76(0.00)	93.72(0.00)	93.84(0.00)	94.00(0.08)	93.80(0.04)	93.84(0.04)	94.06(0.02)	93.98(0.02)	93.56(0.02)	91.79(0.00)
Poker	89.75 (0.05)	89.80(0.01)	89.88(0.03)	89.97(0.02)	90.16(0.03)	90.15(0.16)	90.20(0.03)	90.06(0.04)	89.99(0.47)	90.32(0.47)	94.11(0.00)
Rialto	80.11 (0.11)	79.97(0.07)	79.85(0.14)	79.69(0.00)	79.37(0.07)	79.24(0.09)	79.19(0.05)	78.91(0.00)	78.67(0.10)	78.45(0.10)	78.09(0.00)
AGR	94.67 (0.03)	94.64(0.01)	94.74(0.04)	94.73(0.05)	94.73(0.02)	94.74(0.03)	94.70(0.01)	94.71(0.02)	94.62(0.01)	94.51(0.01)	93.54(0.00)
AGR_a	92.81 (0.11)	92.73(0.15)	93.04(0.03)	92.63(0.45)	93.00(0.11)	92.56(0.24)	92.40(0.36)	92.66(0.02)	92.12(0.17)	91.86(0.17)	90.82(0.00)
AGR_g	89.62 (0.14)	89.92(0.35)	89.56(0.20)	89.68(0.44)	89.72(0.02)	89.76(0.06)	89.49(0.17)	88.43(0.07)	88.23(0.04)	87.51(0.04)	85.93(0.00)
HYPER	85.49 (0.03)	85.62(0.05)	85.46(0.11)	85.49(0.02)	85.34(0.02)	85.21(0.04)	84.85(0.11)	84.46(0.08)	83.88(0.03)	82.87(0.03)	80.96(0.00)
LED	73.96 (0.01)	73.95(0.01)	73.95(0.00)	73.95(0.00)	73.95(0.00)	73.97(0.01)	73.96(0.01)	73.94(0.01)	73.93(0.01)	73.94(0.01)	73.88(0.00)
LED_a	74.03 (0.01)	74.02(0.00)	74.04(0.01)	74.02(0.02)	74.03(0.00)	74.03(0.00)	74.04(0.01)	74.03(0.02)	74.01(0.00)	74.00(0.00)	73.84(0.00)
LED_g	73.24 (0.00)	73.25(0.01)	73.22(0.01)	73.24(0.02)	73.23(0.01)	73.23(0.01)	73.21(0.01)	73.24(0.02)	73.22(0.01)	73.16(0.01)	73.01(0.00)
RBF_f	76.37 (0.03)	76.45(0.01)	76.60(0.01)	76.71(0.00)	76.81(0.01)	76.91(0.05)	77.01(0.01)	77.12(0.04)	77.14(0.02)	76.95(0.02)	75.01(0.00)
RBF_m	86.02 (0.00)	86.01(0.01)	86.04(0.01)	86.05(0.00)	86.05(0.00)	86.09(0.01)	86.10(0.03)	86.16(0.00)	86.25(0.01)	86.39(0.01)	86.07(0.00)
RTG	92.07 (0.43)	91.80(0.02)	90.91(0.05)	91.54(0.05)	90.10(0.63)	90.27(0.51)	91.04(0.37)	90.47(0.16)	90.05(0.59)	88.64(0.59)	85.66(0.00)
SEA	89.77 (0.00)	89.77(0.00)	89.77(0.00)	89.76(0.01)	89.76(0.01)	89.74(0.00)	89.73(0.00)	89.72(0.01)	89.65(0.00)	88.22(0.00)	81.17(0.00)
SEA_a	88.63 (0.16)	88.69(0.15)	88.60(0.02)	88.45(0.17)	88.52(0.05)	88.12(0.12)	87.88(0.28)	86.58(0.48)	86.75(0.28)	83.48(0.28)	81.50(0.00)
SEA_g	87.25 (0.03)	87.21(0.20)	87.41(0.00)	87.12(0.10)	87.21(0.09)	86.71(0.05)	86.24(0.27)	86.16(0.27)	85.70(0.90)	87.50(0.90)	81.08(0.00)
SINE	99.74 (0.00)	99.74(0.00)	99.75(0.01)	99.75(0.01)	99.74(0.00)	99.74(0.00)	99.74(0.00)	99.73(0.01)	99.72(0.01)	99.73(0.01)	99.70(0.00)
WAVEFORM	85.82 (0.00)	85.81(0.01)	85.82(0.01)	85.84(0.02)	85.83(0.01)	85.80(0.00)	85.81(0.01)	85.79(0.01)	85.68(0.00)	85.51(0.00)	85.07(0.00)
WAVEFORM_g	85.47 (0.00)	85.48(0.02)	85.46(0.05)	85.47(0.00)	85.49(0.03)	85.48(0.02)	85.43(0.04)	85.42(0.02)	85.35(0.00)	85.16(0.00)	84.79(0.00)
Ranking médio geral	5,27	5,36	4,91	4,88	4,33	4,85	5,30	5,70	6,52	6,94	8,52

Table 19: Comparison of the Kappa M metric between the SRP algorithm and the application of the PBIS strategy at the classifier level. Bold values represent the best results per metric.

Dataset	SRP	δ									Inc.
		0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	
Airlines	29.42 (0.00)	29.37(0.06)	29.10(0.00)	28.96(0.03)	28.53(0.06)	28.08(0.07)	27.27(0.02)	26.26(0.12)	25.30(0.03)	23.44(0.10)	18.91(0.00)
BankFraud	0.00 (0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.02(0.01)	0.01(0.01)	-2.34(0.00)
Census	23.80 (0.07)	24.33(0.47)	24.93(0.23)	25.28(0.17)	25.96(0.08)	26.69(0.08)	28.00(0.23)	28.96(0.06)	30.09(0.20)	30.95(0.11)	26.40(0.00)
Coil2000	-1.45 (0.28)	-2.30(0.26)	-1.53(0.34)	-1.45(0.09)	-1.50(0.25)	-0.77(0.09)	-1.02(0.00)	-1.53(0.17)	-2.04(0.00)	-4.09(0.17)	-79.52(0.00)
Connect-4	37.82 (0.17)	38.21(0.07)	38.09(0.03)	38.56(0.10)	38.64(0.12)	38.92(0.16)	39.35(0.07)	39.60(0.11)	40.59(0.03)	41.72(0.00)	42.03(0.00)
CovType	90.89 (0.05)	90.91(0.02)	90.95(0.02)	90.97(0.03)	91.01(0.03)	90.98(0.01)	90.98(0.02)	90.94(0.02)	90.98(0.02)	90.92(0.01)	91.34(0.00)
Electricity	75.99 (0.14)	75.76(0.01)	75.59(0.04)	75.60(0.05)	75.74(0.16)	76.16(0.06)	75.97(0.20)	75.85(0.16)	76.66(0.14)	78.44(0.27)	81.86(0.00)
Gassensor	91.29 (0.13)	90.97(0.16)	90.95(0.02)	90.68(0.15)	90.49(0.52)	89.93(0.53)	90.20(0.27)	90.89(0.09)	91.00(0.20)	90.82(0.01)	90.25(0.00)
GMSC	2.69 (0.20)	3.01(0.15)	3.35(0.03)	3.16(0.12)	3.32(0.17)	3.47(0.08)	3.65(0.09)	3.89(0.11)	3.63(0.17)	4.09(0.00)	-4.49(0.00)
Kddcup99	99.74 (0.00)	99.75(0.01)	99.75(0.01)	99.75(0.00)	99.75(0.00)	99.76(0.01)	99.75(0.00)	99.75(0.00)	99.75(0.01)	99.75(0.00)	99.75(0.00)
Keystroke	88.37 (0.41)	87.93(1.07)	88.47(0.13)	88.20(0.00)	88.20(0.47)	87.73(0.34)	87.20(1.01)	86.79(0.20)	87.33(0.06)	86.93(0.88)	84.85(0.00)
NOAA	33.83 (0.00)	34.68(0.15)	34.75(0.20)	34.42(0.13)	34.88(0.27)	35.37(0.15)	34.78(0.12)	34.68(0.27)	34.49(0.22)	33.19(0.04)	28.67(0.00)
Nomao	90.83 (0.00)	90.75(0.09)	90.58(0.06)	90.79(0.01)	90.63(0.14)	90.49(0.02)	90.26(0.13)	90.48(0.01)	90.26(0.09)	90.29(0.02)	91.59(0.00)
Outdoor	67.57 (0.33)	67.97(0.05)	67.53(0.65)	67.71(0.13)	68.50(0.67)	67.45(0.03)	68.06(0.15)	67.97(0.31)	67.70(0.20)	69.14(0.73)	70.00(0.00)
Ozone	-0.62 (1.08)	1.25(0.00)	0.62(0.00)	2.50(0.00)	5.12(0.83)	1.88(0.63)	2.50(0.63)	5.94(0.32)	4.69(0.32)	-1.87(0.63)	-29.99(0.00)
Poker	79.46 (0.11)	79.56(0.01)	79.73(0.05)	79.91(0.04)	80.20(0.08)	80.25(0.31)	80.36(0.06)	80.06(0.09)	79.94(0.94)	80.59(0.13)	88.20(0.00)
Rialto	77.90 (0.12)	77.75(0.08)	77.61(0.15)	77.43(0.01)	77.12(0.07)	76.94(0.10)	76.88(0.06)	76.56(0.01)	76.30(0.11)	76.05(0.13)	75.66(0.00)
AGR	83.75 (0.09)	83.66(0.04)	83.98(0.13)	83.92(0.17)	83.94(0.09)	83.96(0.11)	83.85(0.05)	83.87(0.04)	83.60(0.03)	83.28(0.03)	80.30(0.00)
AGR_a	84.76 (0.23)	84.60(0.32)	85.25(0.07)	84.37(0.96)	84.82(0.38)	84.22(0.52)	83.90(0.76)	84.45(0.04)	83.29(0.36)	82.74(0.05)	80.55(0.00)
AGR_g	78.00 (0.29)	78.63(0.72)	77.88(0.43)	78.12(0.93)	78.08(0.18)	78.30(0.13)	77.72(0.37)	75.49(0.14)	75.05(0.06)	73.52(1.35)	70.18(0.00)
HYPER	70.94 (0.06)	71.19(0.09)	70.87(0.21)	70.94(0.03)	70.56(0.06)	70.37(0.08)	69.64(0.23)	68.87(0.16)	67.72(0.06)	65.69(0.16)	61.86(0.00)
LED	71.04 (0.00)	71.03(0.01)	71.03(0.00)	71.03(0.00)	71.02(0.01)	71.06(0.01)	71.04(0.01)	71.02(0.01)	71.01(0.01)	71.02(0.02)	70.95(0.00)
LED_a	71.09 (0.00)	71.08(0.01)	71.10(0.00)	71.08(0.02)	71.09(0.02)	71.10(0.00)	71.11(0.01)	71.09(0.02)	71.07(0.00)	71.06(0.01)	70.88(0.00)
LED_g	70.22 (0.01)	70.23(0.01)	70.20(0.01)	70.22(0.02)	70.21(0.02)	70.21(0.00)	70.19(0.02)	70.21(0.01)	70.20(0.01)	70.12(0.02)	69.96(0.00)
RBF_f	66.24 (0.04)	66.36(0.01)	66.57(0.01)	66.72(0.00)	66.88(0.01)	67.01(0.06)	67.16(0.00)	67.31(0.05)	67.34(0.02)	67.06(0.07)	64.29(0.00)
RBF_m	80.03 (0.00)	80.00(0.01)	80.06(0.01)	80.07(0.00)	80.07(0.04)	80.13(0.02)	80.14(0.04)	80.23(0.01)	80.35(0.01)	80.55(0.00)	80.10(0.00)
RTG	81.20 (1.02)	80.54(0.07)	78.44(0.13)	79.93(0.13)	77.78(1.46)	76.93(1.22)	78.76(0.88)	77.41(0.38)	76.40(1.41)	73.06(0.21)	65.99(0.00)
SEA	71.34 (0.01)	71.33(0.01)	71.32(0.00)	71.29(0.01)	71.29(0.00)	71.26(0.00)	71.23(0.01)	71.18(0.02)	71.00(0.01)	66.98(0.84)	47.24(0.00)
SEA_a	73.65 (0.37)	73.79(0.34)	73.58(0.04)	73.23(0.39)	73.46(0.15)	72.48(0.27)	71.90(0.66)	68.91(1.11)	69.30(0.64)	61.70(1.91)	57.12(0.00)
SEA_g	70.46 (0.08)	70.35(0.48)	70.82(0.02)	70.15(0.23)	70.07(0.26)	69.20(0.10)	68.10(0.61)	67.93(0.62)	66.86(2.08)	71.03(0.50)	56.16(0.00)
SINE	99.45 (0.00)	99.43(0.00)	99.46(0.02)	99.45(0.01)	99.44(0.00)	99.43(0.00)	99.43(0.01)	99.42(0.02)	99.40(0.02)	99.41(0.01)	99.34(0.00)
WAVEFORM	78.69 (0.00)	78.68(0.02)	78.69(0.02)	78.71(0.02)	78.70(0.01)	78.66(0.00)	78.67(0.01)	78.63(0.01)	78.48(0.01)	78.24(0.02)	77.57(0.00)
WAVEFORM.g	78.18 (0.00)	78.19(0.02)	78.15(0.06)	78.17(0.00)	78.20(0.03)	78.18(0.02)	78.11(0.05)	78.10(0.03)	77.99(0.00)	77.70(0.01)	77.15(0.00)
Ranking médio geral	5,24	5,33	5,36	5,21	4,45	4,97	5,39	5,88	6,58	7,00	8,55

Table 20: Comparison of Kappa M (%) of the original ARF and the versions with instance selection PBIS_E ($\delta = 0.1$) and PBIS_C ($\delta = 0.4$). The comparison column indicates gain (+), loss (-), or tie (=) relative to the original version.

Dataset	ARF	ARF+PBIS _E ($\delta = 0, 1$)		ARF+PBIS _C ($\delta = 0, 4$)	
		Valor	Comp.	Valor	Comp.
Airlines	25,14	24,81	—	24,15	—
BankFraud	-0,02	-0,02	=	0,00	+
Census	24,34	24,75	+	26,40	+
Connect-4	35,66	36,25	+	36,56	+
Coil2000	-2,17	-1,84	+	-1,96	+
CovType	89,68	89,76	+	89,94	+
Electricity	77,84	77,80	—	78,21	+
Gassensor	87,55	86,26	—	86,47	—
GMSC	3,80	3,71	—	3,63	—
Keystroke	82,67	83,53	+	83,67	+
Kddcup99	99,67	99,72	+	99,74	+
NOAA	34,02	33,88	—	33,55	—
Nomao	90,36	90,33	—	90,27	—
Outdoor	62,77	63,25	+	62,48	—
Ozone	-0,15	0,50	+	3,59	+
Poker	77,51	77,85	+	79,04	+
Rialto	69,13	68,72	—	68,19	—
Shuttle	98,79	98,76	—	98,80	+
AGR	81,92	81,96	+	82,74	+
AGR_a	74,24	74,36	+	73,21	—
AGR_g	62,69	63,01	+	62,61	—
HYPER	71,42	71,84	+	71,97	+
LED	70,98	70,96	—	70,99	+
LED_a	71,01	71,01	=	71,02	+
LED_g	70,08	70,10	+	70,05	—
MIXED	99,05	99,05	=	99,05	=
RBF_f	66,11	65,56	—	64,83	—
RBF_m	81,67	81,62	—	81,86	+
RTG	90,73	90,65	—	89,06	—
SEA	71,06	71,01	—	71,00	—
SEA_a	75,58	75,49	—	75,36	—
SEA_g	73,46	74,35	+	74,40	+
SINE	98,97	98,96	—	98,98	+
WAVEFORM	78,14	78,21	+	78,41	+
WAVEFORM_g	77,74	77,79	+	77,93	+
+			17	20	
=			3	1	
-			15	14	

Table 21: Comparison of Kappa M (%) of the original ARTE and the versions with instance selection PBIS_E ($\delta = 0.1$) and PBIS_C ($\delta = 0.4$). The comparison column indicates gain (+), loss (-), or tie (=) relative to the original version.

Dataset	ARTE	ARTE+PBIS _E ($\delta = 0, 1$)		ARTE+PBIS _C ($\delta = 0, 4$)	
		Valor	Comp.	Valor	Comp.
Airlines	26,94	23,41	—	26,20	—
BankFraud	0,00	0,00	=	0,00	=
Census	30,64	32,07	+	33,03	+
Connect-4	36,51	36,19	—	36,81	+
Coil2000	-1,50	-2,45	—	-1,91	—
CovType	90,16	90,79	+	90,96	+
Electricity	79,10	78,12	—	78,65	—
Gassensor	94,24	94,30	+	94,38	+
GSMC	1,00	3,77	+	3,84	+
Keystroke	87,21	86,51	—	87,18	—
Kddcup99	99,48	99,76	+	99,76	+
NOAA	33,34	34,45	+	34,13	+
Nomao	90,90	90,79	—	90,77	—
Outdoor	71,18	72,31	+	72,21	+
Ozone	2,37	3,37	+	2,75	—
Poker	88,84	86,39	—	88,33	—
Rialto	78,64	76,76	—	78,15	—
Shuttle	98,81	98,77	—	98,91	+
AGR	84,73	84,63	—	84,50	—
AGR_a	88,13	88,01	—	87,97	—
AGR_g	82,75	82,43	—	82,56	—
HYPER	75,00	74,10	—	74,71	—
LED	70,53	70,99	+	70,99	+
LED_a	70,00	70,97	+	70,98	+
LED_g	68,76	69,92	+	69,88	—
MIXED	99,29	99,28	—	99,30	+
RBF_f	70,38	70,59	+	70,47	—
RBF_m	82,21	82,35	+	82,76	+
RTG	92,63	92,45	—	92,25	—
SEA	71,59	71,59	=	71,60	+
SEA_a	76,20	76,21	+	76,20	=
SEA_g	75,23	75,21	—	75,20	—
SINE	99,74	99,73	—	99,75	+
WAVEFORM	78,51	78,82	+	78,92	+
WAVEFORM_g	78,05	78,47	+	78,58	+
+			16	20	
=			2	2	
-			17	13	

Table 22: Comparison of Kappa M (%) of the original LB and the versions with instance selection PBIS_E ($\delta = 0.1$) and PBIS_C ($\delta = 0.4$). The comparison column indicates gain (+), loss (−), or tie (=) relative to the original version.

Dataset	LB	LB+PBIS _E ($\delta = 0, 1$)		LB+PBIS _C ($\delta = 0, 4$)	
		Valor	Comp.	Valor	Comp.
Airlines	18,05	17,71	−	17,61	−
BankFraud	-0,19	-0,10	+	-0,14	+
Census	22,21	22,38	+	26,34	+
Connect-4	38,69	38,86	+	36,12	−
Coil2000	-2,30	-2,52	−	-1,94	+
CovType	87,08	87,01	−	83,83	−
Electricity	74,47	74,13	−	77,25	+
Gassensor	74,82	74,34	−	79,82	+
GMSC	3,14	3,31	+	3,29	+
Keystroke	73,25	71,17	−	73,00	−
Kddcup99	99,55	99,55	=	99,55	=
NOAA	31,57	30,18	−	30,61	−
Nomao	87,00	86,61	−	87,78	+
Outdoor	56,31	56,77	−	60,32	+
Ozone	-1,71	-2,99	−	0,37	+
Poker	90,93	89,98	−	64,04	−
Rialto	56,76	56,18	−	55,56	−
Shuttle	96,49	96,37	−	94,48	−
AGR	84,07	84,01	−	83,81	−
AGR_a	78,16	78,02	−	77,89	−
AGR_g	70,54	70,61	+	61,99	−
HYPER	75,07	75,47	+	77,48	+
LED	71,02	71,02	=	71,04	+
LED_a	70,97	71,02	+	71,15	+
LED_g	70,20	70,19	−	70,12	−
MIXED	98,98	98,97	−	99,01	+
RBF_f	42,62	41,92	−	35,59	−
RBF_m	78,70	78,81	+	77,74	−
RTG	94,88	94,79	−	94,45	−
SEA	71,58	71,61	+	71,58	=
SEA_a	73,60	73,82	+	75,77	+
SEA_g	74,27	74,35	+	74,86	+
SINE	99,68	99,70	−	99,77	+
WAVEFORM	78,94	78,96	+	79,10	+
WAVEFORM_g	78,75	78,73	−	78,94	+
+			14	18	
=			4	2	
-			17	15	

Table 23: Comparison of Kappa M (%) of the original SRP and the versions with instance selection PBIS_E ($\delta = 0.1$) and PBIS_C ($\delta = 0.4$). The comparison column indicates gain (+), loss (-), or tie (=) relative to the original version.

Dataset	SRP	SRP+PBIS _E ($\delta = 0, 1$)		SRP+PBIS _C ($\delta = 0, 4$)	
		Valor	Comp.	Valor	Comp.
Airlines	29,42	29,08	—	28,53	—
BankFraud	0,00	0,00	=	0,00	=
Census	23,80	23,87	+	25,96	+
Connect-4	37,82	38,25	+	38,64	+
Coil2000	-1,45	-1,50	—	-1,50	—
CovType	90,89	90,84	—	91,01	+
Electricity	75,99	75,43	—	75,74	—
Gassensor	91,29	91,42	+	90,49	—
GMSC	2,69	3,03	+	3,32	+
Keystroke	88,37	87,88	—	88,20	—
Kddcup99	99,74	99,75	+	99,75	+
NOAA	33,83	34,77	+	34,88	+
Nomao	90,83	90,63	—	90,63	—
Outdoor	67,57	67,73	+	68,50	+
Ozone	-0,62	-0,25	+	5,12	+
Poker	79,46	79,56	+	80,20	+
Rialto	77,90	77,50	—	77,12	—
Shuttle	98,79	98,76	—	98,80	+
AGR	83,75	83,63	—	83,94	+
AGR_a	84,76	85,04	+	84,82	+
AGR_g	78,00	78,64	+	78,08	+
HYPER	70,94	71,27	+	70,56	—
LED	71,04	71,02	—	71,02	—
LED_a	71,08	71,10	+	71,09	+
LED_g	70,21	70,22	+	70,22	+
MIXED	47,08	50,27	+	57,67	+
RBF_f	66,24	66,04	—	66,88	+
RBF_m	80,03	79,96	—	80,07	+
RTG	81,20	81,28	+	77,78	—
SEA	71,34	71,31	—	71,29	—
SEA_a	73,65	74,23	+	73,46	—
SEA_g	70,46	70,67	+	70,07	—
SINE	99,45	99,44	—	99,46	+
WAVEFORM	78,69	78,71	+	78,70	+
WAVEFORM_g	78,18	78,19	+	78,20	+
+			20		21
=			1		1
-			14		13